

LIGHTING SCIENCE: CALCULATION OF LIGHT AND ITS EFFECTS

AN AMERICAN NATIONAL STANDARD

ANSI/IES LS-6-20 (R2025) ERRATA

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Additions will be posted to this list online as they become available. This errata list was last updated **April 5, 2021**.

Additions are shown in **bold face**. Deletions are shown as ~~strikeout~~.

In the next-to-last paragraph of Section 10.2.2 Unified Glare Rating:

The position index of a source, P , is an inverse measure of the relative sensitivity to a glare source at different positions throughout the field of view. Selected values or families of curves were published in early references. The formula for calculating P is given by Equation ~~A-28~~ **A-29** in Annex A – Formulary.

ANSI/IES LS-6-20 (R2025)
+ERRATA 1

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AN AMERICAN NATIONAL STANDARD

Publication of this document
has been approved by IES.
Suggestions for revisions
should be directed to IES.

Prepared for IES
By the Computer Committee



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CONTENTS

1.0	Introduction and Scope	1
1.1	Introduction	1
1.2	Purpose and Scope	1
2.0	The Role and Use of Lighting Calculations	1
2.1	Analysis of Proposed Lighting Systems	2
2.2	Demonstration of Code Compliance	2
2.3	Assessment of Some Aspect of Design Quality	2
2.4	Design of Lighting Equipment	2
2.5	Education	2
3.0	Calculating Illuminance, Luminance, and Flux	3
3.1	Illuminance from Point Sources	3
3.2	Illuminance from Area Sources	4
3.3	Luminance at a Point	4
3.4	Flux on an Area	5
3.5	Approximations	5
3.6	Diffuse Surfaces	6
4.0	Photometric Data for Calculations	8
4.1	Luminaire Photometry for Calculations	8
4.2	Far-Field Luminaire Photometry	8
4.3	Near-Field Luminaire Photometry	9
4.4	Properties of Surfaces and Materials	9
4.4.1	Reflection	9
4.4.2	Transmittance	10
5.0	Model of Light Transport	10
5.1	Direct Component Calculations	11
5.2	Interreflected Component Calculations	11
5.2.1	Radiative Transfer	11
5.2.2	Ray Tracing	13

6.0	Renderings Based on Calculations	15
6.1	Overview of Rendering Generation	15
6.2	Computational Basis	16
6.2.1	Photometric Calculations	16
6.2.2	Radiometric Calculations	16
6.2.3	Radiative Transfer Calculations	16
6.2.4	Ray Tracing Calculations	17
6.2.5	Mixed Calculations	17
6.3	Adding Realism to Renderings	17
6.4	Display Properties and Limitations	18
7.0	Evaluating Lighting Analysis Software	21
7.1	Analytic Tests	21
7.2	Measurement Tests	22
7.3	Testing Software	22
8.0	Factors Affecting Lighting Calculations: Light Loss Factors	22
9.0	Assessing Computed Results	23
9.1	Averages	23
9.2	Minimum and Maximum	23
9.3	Coefficient of Variation	23
9.4	Criterion Ratings	23
10.0	Standardized Calculation Procedures	24
10.1	Calculating Average Illuminance	24
10.1.1	Determining the CU: The Zonal Cavity Method	25
10.1.2	Effective Cavity Reflectances	26
10.1.3	Limitations	26
10.2	Calculating Glare	26
10.2.1	Visual Comfort Probability	26
10.2.2	Unified Glare Rating	27
Annex A – Formulary		27
A.1	Calculating Configuration Factors	29
A.2	Calculating Form Factors	31
A.3	Calculating Lumen Method Coefficients of Utilization	32
A.4	Calculating Luminaire Spacing Criteria	34
A.5	Calculating Solid Angle	36
A.6	Calculating Luminaire Efficiency	37
A.7	Calculating Average Luminaire Luminance	37
A.8	Calculating Visual Comfort Probability (VCP)	38

Annex B – Examples of Application of Radiative Transfer.....40
B.1 Implementation of the Flux Balance Model.....40
B.2 Monte Carlo Ray Tracing.....42

Annex C – Comparison of Calculated and Measured Quantities43

References45

The purpose of computing is insight, not numbers. 1961
The purpose of computing numbers is not yet in sight. 1997
— W. Hamming

1.0 Introduction and Scope

1.1 Introduction

Predicting the performance of a proposed lighting design is an integral part of the design process, allowing the designer to examine and compare alternatives, refine a promising idea, see whether applicable recommendations and codes will be met, evaluate energy conservation and lighting control opportunities, invoke standardized procedures to predict glare and visibility, and perhaps generate a rendering of how a space might appear. The ability to predict performance requires a computational infrastructure that consists of: standardized data that characterize lighting equipment; a knowledge of the properties of surface and other components of the environment involved; theoretical models of how light behaves; software that makes use of those models; and computer hardware on which the software operates. However elaborate this infrastructure, its output still requires careful interpretation.

1.2 Purpose and Scope

The purpose of this Lighting Science document is to provide the theoretical basis for lighting calculations, to describe how this theory is approximated and used, and to describe how it is embodied in most lighting analysis software. This can provide, from a user's perspective, an understanding of the power and limitations of calculations—however performed—and thus make clear their use in the lighting design process.

These purposes require the presentation of information in the following three general areas.

- The fundamental theories of light transport and interaction with architecture, what form—exact or approximate—the mathematical models of these theories take, and how the models are in turn used in lighting software.
- The geometric, photometric, and physical information that is commonly available as input into a lighting calculation process and how the nature, limits, and uncertainties of this information affect the results.
- The various ways of predicting lighting system performance by assessing and interpreting calculation results and their comparison with anticipated or actual measured results.

In addition, this document explains IES standard calculation procedures, including lumen method coefficients of utilization and glare assessment.

2.0 The Role and Use of Lighting Calculations

Like most technical disciplines in engineering and design, lighting had a long history of support and direction provided by hand calculation, nomograms, and mechanical calculators.¹ By the middle of the 20th century these were, to some extent, augmented with large analog and then digital electronic computers. But the eventual widespread availability and use of inexpensive computers running standardized operating systems has made general purpose lighting analysis software (hereafter simply “software”) a commercially viable enterprise, and hand calculations, however augmented, now have virtually no role in modern illuminating engineering or lighting design.

Software is available from independent developers, lighting equipment manufacturers, equipment development consultants, and government agencies,² and now has a part, however small, in the design and specification of most lighting systems. The development of general purpose software is a vast, highly technical undertaking that requires knowledge not only of illuminating engineering and lighting design but also mathematics and computer programming.³⁻⁶ In almost all cases, the complexities of the calculation process, including the fundamental basis and the assumptions that are always part of any practical calculation, are expressly and appropriately hidden from the user in the course of software usage. Nevertheless, some knowledge of these assumptions and approximations, coupled with an experience-based knowledge of the likely range of a quantitative result, helps prevent software misuse or misapplication and detects bad or inappropriate input data.

Reliability of software is usually established by comparison with simple geometric and photometric settings that can be calculated by hand from first principles or by comparison with photometric measurements. (See **Section 7.0 Evaluating Lighting Analysis Software**.) The presumed and demonstrated reliability of software permits it to be used in several important aspects of illuminating engineering and lighting design.

2.1 Analysis of Proposed Lighting Systems

Software is used most frequently to demonstrate that a proposed lighting system produces illuminances that meet recommendations in their various forms and from various recommending or governing bodies. Luminance limits, luminance ratios, and proper luminaire placement can also be efficiently investigated with software.

In addition to performance testing, software can determine the effect of inevitable uncertainties in building or environmental parameters, such as surface reflectances and furniture placement.

2.2 Demonstration of Code Compliance

Code compliance involving illuminance minimums in interiors, various maximums in exterior applications, and lighting power density limits are usually demonstrated with software.

2.3 Assessment of Some Aspect of Design Quality

The advent of sophisticated computer graphics rendering capabilities provides another use for software in the lighting design process. Renderings that are photometrically accurate and photographically realistic provide feedback to the lighting designer that his or her design will have the intended “look” in addition to meeting quantitative criteria. They also permit the communication and verification of lighting design ideas to clients and other design team members. Though limited by the capability of computer display technology, such renderings can help provide clearer conceptions of lighting system effects and performance.

2.4 Design of Lighting Equipment

Software has had a significant effect on the design of lighting equipment. The ability to accurately predict proposed luminaire performance is now used by equipment manufacturers to shorten the design-cycle time, reduce development costs for new product development, and permit quicker and more extensive investigation of equipment design concepts. The repeated cycle of physical mockups and photometric testing has been significantly reduced, and is now used only near the end of the luminaire design process.

2.5 Education

Software permits students of lighting to learn and explore lighting concepts, test design ideas, and add to their store of experience with light, lighting, and lighting equipment. Though no substitute for real, hands-on work, software can significantly augment the range and depth of understanding of lighting system performance and provide an initial experience with a range of lighting equipment and lighting techniques that is otherwise very difficult to obtain.

3.0 Calculating Illuminance, Luminance, and Flux

All computational assessments of lighting systems are based on the determination of illuminance, luminance, or flux. This is the case for everything from the simplest determination of average illuminance on a work plane to the most elaborate computer rendering of a lighted space. All of these calculations involve:

- Photometric properties of light sources
- Surface and material properties, including reflectance, transmittance, refraction, and color
- The geometry defining and relating sources and surfaces
- A final form of the computation, which can range in complexity from an array of illuminance values to a rendered image

Virtually all lighting calculations are performed with the assumption that the air through which light travels is clear and non-absorbing. The atmospheric conditions that would seem to absorb light and affect daylighting calculations are taken into account when the sun and sky are defined as light sources.

Calculation of illuminance, luminance, and flux can be expressed in rigorously correct terms as equations derived from first principles. Though of theoretical interest, these equations are usually approximated for practical implementation. Certain calculations can be simplified and less computer time used if the sources involved are perfectly diffuse emitters or if the surfaces involved exhibit perfectly diffuse reflection.

3.1 Illuminance from Point Sources

The most fundamental and conceptually simplest calculation is the determination of the illuminance at a point produced by a point source. For the purposes of this discussion, it is assumed that the intensity distribution of the source is defined by a function $I(\theta, \psi)$ with direction specified in spherical coordinates (θ, ψ) centered at the source. Assuming the geometric arrangement shown in **Figure 3-1**, the illuminated area dA_p located at point p is in direction (θ, ψ) from the source and distance D from it. The orientation of the illuminated surface is indicated by the surface perpendicular, n .

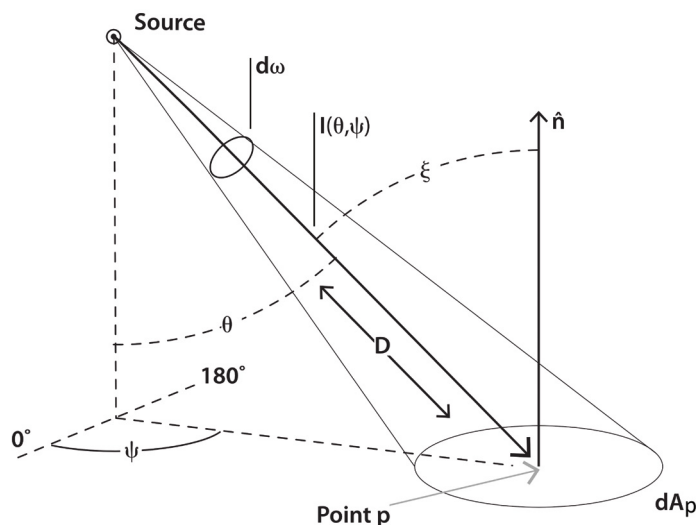


Figure 3-1. Geometry of the calculation of illuminance at a point from a point source. (© Illuminating Engineering Society)

Beginning with the definition of illuminance, and using the definition of luminous intensity, an equation for the illuminance at point p can be derived:

$$E_p = \frac{d\Phi}{dA_p} = \frac{I(\theta, \psi) d\omega}{dA_p} = \frac{I(\theta, \psi) dA_p \cos \xi}{dA_p D^2} = \frac{I(\theta, \psi) \cos \xi}{D^2} \quad (3-1)$$

This is the so-called inverse-square cosine law. Since dA_p is differentially small, it is assumed that neither $I(\theta, \psi)$, nor the distance, nor any of the angles change for any point in the differential neighborhood of dA_p , and so the illuminance is the same within dA_p .

3.2 Illuminance from Area Sources

When the source is a luminous area, it is convenient to describe it photometrically in terms of luminance rather than intensity. The luminance distribution of the source is defined by a function $L(\theta, \psi; u, v)$ with direction specified in spherical coordinates (θ, ψ) centered at a point on the source located at (u, v) . **Figure 3-2** shows the arrangement.

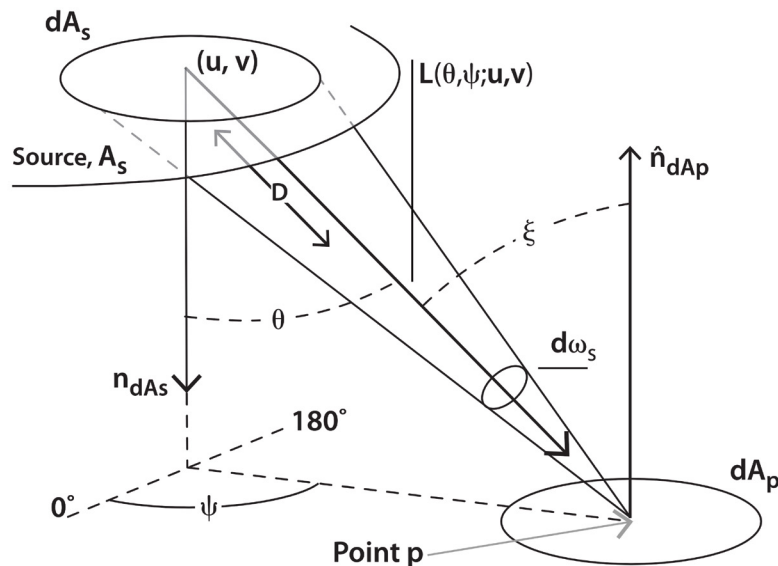


Figure 3-2. Geometry of the calculation of illuminance at a point from an area source. (© Illuminating Engineering Society)

Beginning with the definitions of illuminance and luminance, an expression for the differential illuminance at point p due to the luminance of a differential element of source can be found. This is integrated (summed) over the entire source to give the illuminance at point p produced by the entire source:

$$E_p = \int dE_p = \int d\frac{d\Phi}{dA_p} = \int \frac{dI_{dA_p}(\theta, \psi) d\omega_s}{dA_p} = \int \frac{L(\theta, \psi; u, v) dA_s \cos\theta dA_p \cos\xi}{dA_n D^2} = \int L(\theta, \psi; u, v) \cos\xi d\omega_s \quad (3-2)$$

3.3 Luminance at a Point

The luminance at a point p on a surface is calculated from the illuminance at that point, integrated with the directional reflectance of the surface. The BRDF of the surface, $f_r(\theta_i, \psi_i; \theta_r, \psi_r)$, characterizes the directional reflectance. (See ANSI/IES LS-2-20, *Lighting Science: Concepts and Language of Lighting*.⁷) The luminance in direction (θ_r, ψ_r) is given by:

$$L_p(\theta_r, \psi_r) = \int f_r(\theta_i, \psi_i; \theta_r, \psi_r) dE(\theta_i, \psi_i), \quad (3-3)$$

where:

$dE(\theta_i, \psi_i)$ = differential amount of illuminance at the point from a direction (θ_i, ψ_i)

Using **Equation 3-2**, L_p can be expressed explicitly as a function of the luminance distribution, $L_s(\theta, \psi)$, of the environment with respect to point p, substituted in **Equation 3-3**. The luminance at point p becomes:

$$L_p(\theta_r, \psi_r) = \int_{\Omega_s} f_t(\theta_i, \psi_i; \theta_r, \psi_r) L_s(\theta_i, \psi_i) \cos \xi d\omega_s \quad (3-4)$$

It should be noted that **Equation 3-4** holds for the use of the BTDF of materials as well, and thus for the calculation of the transmitted luminance of a material.

3.4 Flux on an Area

Flux determination is a critical part of most calculations that involve interreflected light. The flux incident on a surface is obtained by multiplying the illuminance at a point by the differential area around that point, and integrating over the entire receiving area. If the source is a point, then the flux it produces on an area, A_r , is given by:

$$\Phi = \int_{A_r} \frac{I(\theta, \psi) \cos \xi}{D^2} dA_r, \quad (3-5)$$

where:

A_r = entire area of the receiver

If the source is an area, then the flux it produces on another area, A_r , is given by:

$$\Phi = \int_{A_r} \int_{\Omega_s} \frac{I(\theta, \psi) \cos \xi}{D^2} d\Omega_s dA_r, \quad (3-6)$$

where:

Ω_s = entire solid angle of the source

A_r = entire area of the receiver

3.5 Approximations

As a practical matter, a continuous function of luminous intensity is never available to describe lighting equipment. Rather, an array of discrete values, $I'(\theta_i, \psi_j)$, $i=1, \dots, N$, $j=1, \dots, M$, that results from photometry is required. In this common case it is necessary to use an interpolated value of luminous intensity $I'(\theta_i, \psi_j)$. An interpolation procedure is used to generate the value $I'(\theta, \psi)$ from bracketing values available in the array of discrete values: $I'(\theta, \psi_j)$, $I'(\theta_{i+1}, \psi_j)$, $I'(\theta_i, \psi_{j+1})$, $I'(\theta_{i+1}, \psi_{j+1})$. The details of the procedure can have a large effect on the final value, especially if the intensity is changing rapidly in θ or ψ in the direction of point p. Similarly, a continuous function of luminaire luminances is almost never available, and luminances need to be inferred from values of luminous intensity and luminous area.

The integrals indicated in **Equations 3-2** through **3-6** can rarely be performed analytically to obtain a closed-form expression. Rarely are all the functions and their integrands available in analytic form, and the geometric settings required in practice usually involve relationships to the variables of integration that are overwhelmingly complex. Thus, in nearly all cases, numerical integration has to be used rather than analytic integration. This can be accomplished by discretizing areas into small elements, evaluating integrands for each of these elements, and summing the resulting values. An alternative is to transform the area integrals into contour or edge integrals, discretize the edges, and numerically integrate around the edges of the source.⁸ This approximation has been shown to be computationally faster than area discretization with no loss in accuracy.

The final result depends on the granularity of the discretization. In many cases, it is not possible to determine in advance how fine the discretization needs to be to provide useful accuracy. It is often necessary to iterate or pass through the discretization-calculation cycle more than once, increasing the discretization until the results do not change in a significant way. This general procedure is called *adaptive computation*. In some software, this process is automatic; in others, it is under user control.

3.6 Diffuse Surfaces

A surface that emits flux in a way such that it exhibits an intensity distribution that varies with the cosine of the exitant angle measured from the surface perpendicular is said to have a perfectly diffuse intensity distribution. This is usually shortened to “diffuse distribution.”

The intensity perpendicular to the surface, I_n , is the largest in the distribution, and:

$$I(\theta) = I_n \cos\theta$$

where:

θ is the exitant angle measured from the perpendicular

It should be noted that the distribution is not dependent on the azimuthal angle. The simplicity of the function describing a diffuse distribution permits far-reaching simplifications in the equations involving diffuse surfaces as light sources or as elements in a system of interreflecting surfaces. This is particularly important since many real surfaces are approximately diffuse, and calculations involving them can be radically simplified, leading to software that executes rapidly and so can have a significant place in the lighting design and analysis process.

A surface can be diffuse because it is a perfectly diffuse reflector or a perfectly diffuse transmitter, or because it generates light in a perfectly diffuse manner. Many matte painted architectural surfaces are reasonable approximations of diffuse reflectors. Diffusing skylight domes are reasonable approximations of diffuse transmitters. Many OLED sources have a distribution that is close to diffuse.

The following basic properties of the photometric characteristics of a diffuse surface are a direct result of its simple distribution.⁹

$$I(\theta) = I_n \cos\theta$$

$$I_n = \frac{\Phi_{\text{off}}}{\pi} = \Phi_{\text{on}} \frac{\rho_{\text{pdr}}}{\pi} = EA \frac{\rho_{\text{pdr}}}{\pi} = \frac{MA}{\pi} \quad (3-7)$$

$$L(\theta, \psi) = \frac{I(\theta)}{A \cos\theta} = \frac{I_n \cos\theta}{A \cos\theta} = \frac{I_n}{A}$$

$$L = \frac{I_n}{A} = \frac{MA}{\pi A} = \frac{M}{\pi}$$

Using these properties and the definition of solid angle, **Equation 3-2** can be simplified to give the illuminance at a point from a uniform, diffuse area source:

$$E_p = \frac{M}{\pi} \int_{\Omega_s} \frac{\cos\xi}{D^2} d\omega_s = M \frac{1}{\pi} \int_{A_s} \frac{\cos\theta \cos\xi}{D^2} dA_s \quad (3-8)$$

The diffuse assumption permits the photometric properties of the surface to be expressed as a single value of exitance, M , and to be completely separated from the surface's geometric properties. That is, the integral is a purely geometric quantity. It is called the configuration factor, c .¹⁰ Thus, if:

$$c = \frac{1}{\pi} \int_{A_s} \frac{\cos\theta \cos\xi}{D^2} dA_s, \quad (3-9)$$

then the illuminance at a point from a uniform diffuse surface with exitance M_s is simply:

$$E_p = M_s c \quad (3-10)$$

If the direction of the flux flow assumed in **Figure 3-2** is reversed and the area at point p is assumed the diffuse source, then an expression for the fraction of the total flux it emits that directly reaches the large area is also given by **Equation**

3-9. Therefore, a configuration factor can also be defined as that fraction of the total flux emitted by a differential diffuse emitter that is received directly by an area. This shows that the configuration factor has the limiting values $0 \leq c \leq 1$.

Equation 3-9 can be analytically evaluated for a large number of geometric conditions to produce closed-form expressions for the configuration factor.¹¹ A selection of these expressions is in **Annex A – Formulary**. The purely geometric nature of the configuration factor permits the development of what is called *configuration factor algebra* as well as a geometric analogy of its value. **Figure 3-3** shows this analogy.

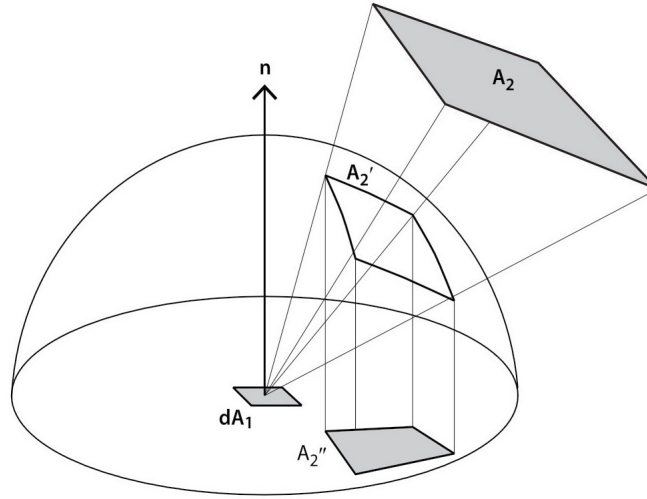


Figure 3-3. The Nusselt analogy for the computation of a configuration factor for surface A_2 and the point at dA_1 . Surface A_2 is radially projected onto a hemisphere with base centered at dA_1 , giving projection A_2' . This, in turn, is projected downward to the base of the hemisphere, giving projection A_2'' . The configuration factor is numerically equal to the area of A_2'' divided by the area of the circular base of the hemisphere.

The analogy also shows another remarkable property of uniform diffuse sources: the illuminance they produce at a point depends only on their outline or silhouette as seen from the point. This property can also be demonstrated mathematically.¹²

If a diffuse area source is considered to illuminate an area receiver, then **Equation 3-6 (Section 3.4)** simplifies, and the flux $\Phi_{s \rightarrow r}$ from a uniform diffuse area source A_s with uniform exitance M_s , that directly reaches a receiving area A_r is:

$$\Phi_{s \rightarrow r} = \frac{M_s}{\pi} \int_{A_r} \int_{A_s} \frac{\cos \theta \cos \xi}{D^2} dA_s dA_r \quad (3-11)$$

This assumes that all points on A_s have an unobstructed view of A_r . As with the configuration factor, the integral is a purely geometric quantity. It is customary to scale the geometric quantity by the source area. Therefore:

$$\Phi_{s \rightarrow r} = \frac{A_s M_s}{\pi} \frac{1}{A_s} \int_{A_r} \int_{A_s} \frac{\cos \theta \cos \xi}{D^2} dA_s dA_r = \Phi_s \frac{1}{\pi A_s} \int_{A_r} \int_{A_s} \frac{\cos \theta \cos \xi}{D^2} dA_s dA_r \quad (3-12)$$

Analogous to the configuration factor, a form factor $f_{s \rightarrow r}$ can be defined as the purely geometric part of **Equation 3-12**:

$$f_{s \rightarrow r} = \frac{1}{\pi A_s} \int_{A_r} \int_{A_s} \frac{\cos \theta \cos \xi}{D^2} dA_s dA_r \quad (3-13)$$

The form factor $f_{s \rightarrow r}$ gives the fraction of flux leaving A_s that gets directly to A_r . Thus:

$$\Phi_{s \rightarrow r} = \Phi_s f_{s \rightarrow r} \quad (3-14)$$

Equation 3-13 can be analytically evaluated for a large number of geometric conditions to produce closed-form expressions for the form factor.¹⁰ A selection of these expressions is in **Annex A – Formulary**. Like configuration factors, form factors can be shown to obey an algebra, and among the relationships that can be established are:

$$\begin{aligned} 0 &\leq f_{s \rightarrow r} \leq 1 \\ \sum_{r=1}^N f_{s \rightarrow r} &= 1 \text{ (for a completely enclosed environment)} \\ A_s f_{s \rightarrow r} &= A_r f_{r \rightarrow s} \\ f_{r \rightarrow s} &= \frac{A_s}{A_r} f_{s \rightarrow r} \end{aligned} \quad (3-15)$$

Thus, for example, the average illuminance at A_r produced by uniform diffuse source A_s is:

$$\bar{E}_r = \frac{\Phi_{s \rightarrow r}}{A_r} = \frac{\Phi_s f_{s \rightarrow r}}{A_r} = \frac{\Phi_s A_r}{A_r A_s} f_{r \rightarrow s} = \frac{\Phi_s}{A_s} f_{r \rightarrow s} = \bar{M}_{s, f_{r \rightarrow s}} = \bar{E}_s \rho_s f_{r \rightarrow s} \quad (3-16)$$

Configuration and form factors form the basis for the diffuse radiative transfer analysis that is incorporated into most software used in lighting design and analysis.

4.0 Photometric Data for Calculations

Lighting calculations involve the photometric characteristics of luminaires and architectural materials, quantified and supplied in a form that can be used in commonly available lighting software. Measurement procedures for acquiring data describing luminaire photometry and surface reflectance and transmittance have been standardized to permit such use.

The reader is also directed to *ANSI/IES LM-72-20, Approved Method: Directional Positioning of Photometric Data* for information on conventional means of describing luminaire orient, tilt, roll, and spin.¹³

4.1 Luminaire Photometry for Calculations

Equivalent luminous intensity distributions are used to specify the spatial flux distribution characteristics of luminaires. (Refer to *ANSI/IES LS-2-20⁷* and *ANSI/IES LS-4-20, Lighting Science: Measurement of Light – The Science of Photometry*.¹⁴) Luminaire photometry gives the equivalent luminous intensity in a set of directions, with the angular spacing of measurements sufficiently small to provide an accurate and useful description of the distribution.

4.2 Far-Field Luminaire Photometry

Photometry determines intensity by calculation, using an illuminance measurement and a test distance. For most photometry, the test distance is constant for all measurement points. Thus, illuminance measurements are made at positions on an imaginary sphere with a radius equal to the test distance. The sphere center coincides with a fiducial point (a point of reference) inside the luminaire. This so-called *photometric center* is often the origin of the coordinate system used for calculations. If illuminances are calculated from these data, correct values result only if the calculation distance is the same as the photometric test distance and the illuminated surface's perpendicular is oriented toward the photometric center of the luminaire, setting aside all other factors that might affect the result.

However, regardless of the luminaire size, it is always possible to choose a photometric distance, D_t , sufficiently large that illuminances produced at distances greater than D_t vary (nearly) as the inverse-square of the distance to the photometric center. This was first shown for diffuse emitters.^{15,16} For them, if D_t was five times the maximum dimension of the emitter, a computational accuracy of at worst 2% resulted. This “five times rule” has been adopted as standard photometric practice.¹⁷ This is far-field photometry, and a distance of at least D_t is used to make the illuminance measurements from which the equivalent luminous intensities are calculated. These intensities can then be used to calculate illuminances at distances greater than D_t , treating the luminaire as a point source. Virtually all commercial photometry is far-field photometry.

Illuminance calculations at distances less than D_t and which assume the luminaire to be a point source are likely to be inaccurate.^{18,19} It should be noted that distributions other than diffuse have different values of D_t ; however, it is customary to apply the five-times rule to most indoor luminaire photometry.

4.3 Near-Field Luminaire Photometry

Near-field photometry describes the spatial flux distribution of a luminaire in a manner permitting accurate illuminance calculations at distances less than D_t . Near-field photometry is particularly important for analyzing indirect lighting systems. Two types of near-field photometry have been developed expressly for improving computational accuracy.

Application-distance photometry uses test distances that are equal to the distances at which illuminance calculations will be made.²⁰⁻²² No assumptions about distance invariance are made. In this case, the luminaire is treated as a point source for calculations. Since illuminance calculations are likely to be made at many distances, application-distance photometry provides intensity distributions for several test distances.

Luminance-field photometry measures and reports the luminance distribution of the luminaire as viewed from a set of points surrounding the luminaire.^{23,24} All points are the same distance from the luminaire photometric center. Precisely stated, the data describe a four-dimensional scalar field of luminance. From these luminance data, illuminance can be calculated at any distance and orientation from the luminaire. Luminance-field photometric measurements can be made using a CCD video camera;¹⁹ however, the quantity of data can be difficult to manage.⁵

4.4 Properties of Surfaces and Materials

In many cases, the reflectance or transmittance of surfaces is not known at design time, and it is often assumed that this limits, if not eliminates, the utility of calculations. However, in such a case, if calculations are performed with the lowest and highest values of reflectance or transmittance that can be reasonably expected, the resulting set of calculations reveals the sensitivity of lighting system performance to surface finishes.

4.4.1 Reflection. Many surfaces and finishes used in architecture exhibit a reflectance that is sufficiently diffuse to be considered perfectly diffuse for design and calculation purposes. This is important, since they can be considered diffuse emitters regardless of the incident direction of the light. Unless otherwise expressly stated, most software assumes that the reflectances specified by the user are perfectly diffuse.

In some cases, assumptions about diffuseness will lead to very inaccurate results. An example of this is the calculation of the luminance of visual tasks, such as pencil marks on paper, etched marks on a rule, and roadway surfaces. The bidirectional-reflectance distribution function (BRDF; refer to *ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering*²⁵) of a surface is used in these cases. Modeling luminaire performance and generating photorealistic renderings of lighted environments require bidirectional reflectance information about surfaces.²⁶⁻²⁸

In addition to simplifying the spatial distribution characteristics of reflectance, it is often permissible and necessary to simplify its spectral characteristics. It is a useful approximation to assume that reflectance is spectrally flat, and it is then assigned a “gray reflectance” equal to the integrated value of the surface’s spectral reflectance illuminated by an equal energy source. Thus, it is assumed that the surface exhibits the same reflectance regardless of the spectral power distribution of incident light used in calculations. This is referred to as the “gray assumption.”

Although this simplifies calculations and usually introduces only small errors, difficulties can arise. The reflectance of a surface with a deeply saturated color usually has a significant reflectance only in a narrow band of wavelengths. Use of a light source spectrally different from that used in making a reflectance measurement can then lead to large quantitative errors and significantly shifted rendered color. Errors can also arise when calculating with broadband “white” light and saturated colored surfaces that are assumed gray for the purpose of the calculations.²⁹

4.4.2 Transmittance. Many transmissive surfaces used in architecture are either image-preserving or diffuse. (Refer to *ANSI/IES LS-3-20, Lighting Science: Physics and Optics of Radiant Power*.³⁰) These two cases can usually be treated with sufficient accuracy, if the gray assumption is made (i.e., the medium is spectrally flat; see **Section 4.4.1 Reflection**), with the transmittance represented by a single value. Some software treats transmittances this way, while other software does not make the gray assumption and uses either full spectral transmittance data or an approximation using wide-band red, green, and blue transmittances.

In some cases, assumptions about single-value transmittance will lead to very inaccurate results. Examples are the large change in transmittance of ordinary window glass at high incident angles and the transmittance of daylight delivery fenestration systems designed to redirect and disperse sunlight. In the first case, an angular transmittance function of one angle suffices; in the second, the bidirectional transmittance distribution function (BTDF) of the material is used.^{31,32} (See **Annex A – Formulary** and ANSI/IES LS-3-20 for equations and useful approximations.)

5.0 Model of Light Transport

Most of the quantitative characteristics of a proposed lighting design can be determined from the calculation of illuminance, sometimes augmented with information about incident directions. All of the most basic assessments of lighting systems use illuminance. These include:

- illuminance at an array of points
- illuminance averages
- illuminance ratios involving averages, minimums, or maximums

Surface luminances can be determined from detailed illuminance information and the reflection or transmission properties of the surface. These cases include:

- luminances to determine visual target contrast
- luminances in glare assessments
- luminances of roadways
- luminances to build a rendering of a lighted space

In virtually all lighting calculations, it is conceptually and computationally convenient to separate the calculation of illuminance into direct and interreflected components. The direct component is the flux incident directly from a

source, i.e., a luminaire or a component of a daylight delivery system. The interreflected component is the flux incident from surfaces made luminous by the multiple reflections of light within a space.

The total illuminance is the sum of the two components:

$$E = E_{\text{direct}} + E_{\text{interreflected}} \quad (5-1)$$

Some assessments have only a direct component, such as the calculations of either illuminance or luminance performed in many outdoor, roadway, and sports applications, where the interreflected component is presumed negligible or is required to be ignored. Some assessments have only an interreflected component, such as work plane illuminance calculations in a space with only an indirect lighting system. Most assessments of interior lighting applications, and some outdoor applications, have both direct and interreflected components.

For computation of maintained values, light loss factors are used in the calculation of both the direct and interreflected components of **Equation 5-1**.

5.1 Direct Component Calculations

Direct illuminance component calculations from luminaires are driven by the available photometric and geometric information. In virtually all cases, only far-field photometric information is available, and it is reported in one of the standard electronic formats.³³ This very basic photometric information requires additional assumptions if it is used to predict the effects of real luminaires. (See **Section 3.0 Calculating Illuminance, Luminance, and Flux**, and **Section 4.1 Luminaire Photometry for Calculations**.)

Direct-component calculations from daylight delivery systems are driven by models of the sun and sky as light sources, and available photometric and geometric information about glazing, louvers, shades, light shelves, and other devices used in daylighting. (Refer to *ANSI/IES LP-3-20, Lighting Practice: Designing and Specifying Daylighting for Buildings*.³⁴)

5.2 Interreflected Component Calculations

The determination of the interreflected illuminance component requires the luminance of surfaces that comprise the lighted environment. Generally, there are two methods used to determine these: radiative transfer and ray tracing. In computer graphics literature the interreflected component is described as the “global illumination.”

5.2.1 Radiative Transfer. Radiative transfer uses the bulk flux model of light transport to perform lighting calculations. Light transport is modeled as a divergent flux beam that radiates outward from a surface. Radiant or luminous flux is introduced into a radiative transfer model of an architectural environment by luminaires or daylight sources. This flux reaches various surfaces, where it is reflected and scattered as additional divergent flux beams or is absorbed. These reflected divergent flux beams then radiate to other surfaces. The process can be continued multiple times, accounting for as many multiple reflections as need requires or resources permit. At the end of this process the total number of lumens arriving at a surface, combined with its reflectance, can be used to determine luminance.

Radiative transfer computation is radically simplified if all surfaces involved are assumed to exhibit perfectly diffuse reflectance. That is, reflected flux has a diffuse distribution, independent of incident direction, and so the transfer between surfaces is simple to express. Most radiative calculations in architectural lighting make this assumption. In computer graphics literature the diffuse radiative transfer method is referred to as “radiosity.”

Inherent in this model is the treatment of radiatively interacting surfaces as collections of discrete elements that exhibit constant reflective, transmissive, and photometric properties across their extent. A flux balance equation

is written for each element, equating the total flux leaving an element to the total incident flux multiplied by the element's reflectance.

The total incident flux has a direct component due to electric and daylight sources, and an interreflected component due to the flux from all the other elements. The equality expressing the flux balance exists when all interreflections are taken into account. For the i^{th} element of a radiative transfer system, the equation is:

$$\Phi_i = \Phi_{0i} + (f_{1 \rightarrow i} \Phi_1 \rho_1 + f_{2 \rightarrow i} \Phi_2 \rho_2 + \dots + f_{m \rightarrow i} \Phi_m \rho_m) \quad (5-2)$$

This can be written for each surface element in the system $i=1, \dots, i=m$, and a set of linear, independent, simultaneous equations results. Expressed in matrix form, this gives:

$$\begin{bmatrix} \Phi_1 \\ \Phi_2 \\ \vdots \\ \Phi_m \end{bmatrix} = \begin{bmatrix} \Phi_{01} \\ \Phi_{02} \\ \vdots \\ \Phi_{0m} \end{bmatrix} + \begin{bmatrix} \rho_1 f_{1 \rightarrow 1} & \rho_1 f_{2 \rightarrow 1} & \dots & \rho_m f_{m \rightarrow 1} \\ \rho_1 f_{1 \rightarrow 2} & \rho_2 f_{2 \rightarrow 2} & \dots & \rho_m f_{m \rightarrow 2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_1 f_{1 \rightarrow m} & \rho_2 f_{2 \rightarrow m} & \dots & \rho_m f_{m \rightarrow m} \end{bmatrix} \times \begin{bmatrix} \Phi_1 \\ \Phi_2 \\ \vdots \\ \Phi_m \end{bmatrix}, \quad (5-3)$$

where:

m = number of elements in the system

Φ_i = flux onto the i^{th} element, due to direct and interreflected flux

ρ_i = diffuse reflectance of the i^{th} element

Φ_{0i} = direct flux onto the i^{th} element (due to luminaires and daylight sources)

$f_{i \rightarrow j}$ = form factor from element i to element j , accounting for occluding surfaces

The above matrix equation can also be rewritten in terms of the illuminance on each surface. This form is particularly valuable when one or more of the reflectances is assigned a value of zero, since then the illuminance striking each element is independent of the element's reflectance. It should be noted that this transformation changes the order of the subscripts of the form factors. Thus:

$$\begin{bmatrix} E_1 \\ E_2 \\ \vdots \\ E_m \end{bmatrix} = \begin{bmatrix} E_{01} \\ E_{02} \\ \vdots \\ E_{0m} \end{bmatrix} + \begin{bmatrix} \rho_1 f_{1 \rightarrow 1} & \rho_2 f_{2 \rightarrow 1} & \dots & \rho_m f_{m \rightarrow 1} \\ \rho_1 f_{1 \rightarrow 2} & \rho_2 f_{2 \rightarrow 2} & \dots & \rho_m f_{m \rightarrow 2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_1 f_{1 \rightarrow m} & \rho_2 f_{2 \rightarrow m} & \dots & \rho_m f_{m \rightarrow m} \end{bmatrix} \times \begin{bmatrix} E_1 \\ E_2 \\ \vdots \\ E_m \end{bmatrix} \quad (5-4)$$

Equation 5-4 can be written more compactly as:

$$\vec{E} = \vec{E}_0 + \vec{F} \vec{\rho} \vec{E}, \quad (5-5)$$

where:

$\vec{E}$ = vector of final illuminances

$\vec{E}_0$ = vector of initial (direct) illuminances

$\vec{F}$ = matrix of form factors

$\vec{\rho}$ = diagonal matrix of diffuse reflectances

The simplest way to solve large matrix systems is to use the method of iteration.^{5,35} The iteration begins by setting the vector $\vec{E}$ equal to the direct illuminance vector $\vec{E}_0$ and using it as an initial estimate of vector $\vec{E}$ after interreflections. This initial estimate is then used on the right-hand side of **Equation 5-5** to generate another estimate of the solution vector on the left. It can be shown that the process converges. Various other iteration methods are used that speed up convergence and use less computer memory. Some geometrically simple systems permit closed-form solutions. (See **Table A-1** in **Annex A – Formulary**.)

Finally, the interreflected component of illuminance at a point is obtained by applying **Equation 3-10** (see **Section 3.6**) to each interreflecting surface in the system, using the illuminance at those surfaces found by solving **Equation 5-5**, and adding the individual contributions:

$$E_{\text{inter}} = \sum_{i=1}^N M_i c_i = \sum_{i=1}^N E_i \rho_i c_i, \quad (5-6)$$

where:

- E_{inter} = the interreflected component of the illuminance
- c_i = configuration factor from p to surface element i , accounting for occlusion
- ρ_i = diffuse reflectance of surface element i
- E_i = illuminance of surface element i , accounting for all interreflections

Generally, discrete radiative transfer models use the “gray assumption” (see **Section 4.4.1 Reflection**) for reflectances and the total lumen output of luminaires to solve for spectrally flat illuminances at each element. If surface reflectances represent fairly saturated colors, or if sources are strongly colored, then spectral power distributions of luminaires and spectral reflectances of surfaces can be discretized into wavelength bands, and the general problem solved for each wavelength band. These multiple solutions provide a spectral distribution of illuminances at each surface element. Alternatively and less accurately, the radiative transfer problem can be solved using three very broad bands in the spectrum, corresponding to the SPDs of three primaries in the RGB color system.

Discrete radiative transfer models can assign each surface only one reflectance, one transmittance, and a single value of incident flux. If it is known that a surface exhibits changing reflectance across its extent, accurate modeling requires that the surface be broken into smaller pieces (discretized) according to its distribution of reflectances. These smaller pieces are usually three- or four-sided polygons. The network of vertices and borders of this collection of polygons is called a *mesh*.

Further, if it is anticipated that a surface will exhibit changing luminance or color across its extent, then it is critical that it be meshed according to these changes in order for the final calculation to be sufficiently accurate. That is, small elements are needed where the changes are large, and large elements suffice where the changes are small. In this sense, the fidelity of the result depends on the nature and granularity of surface meshing.

Figure 5-1 shows meshing for a simple rectangular room containing shadowing objects.

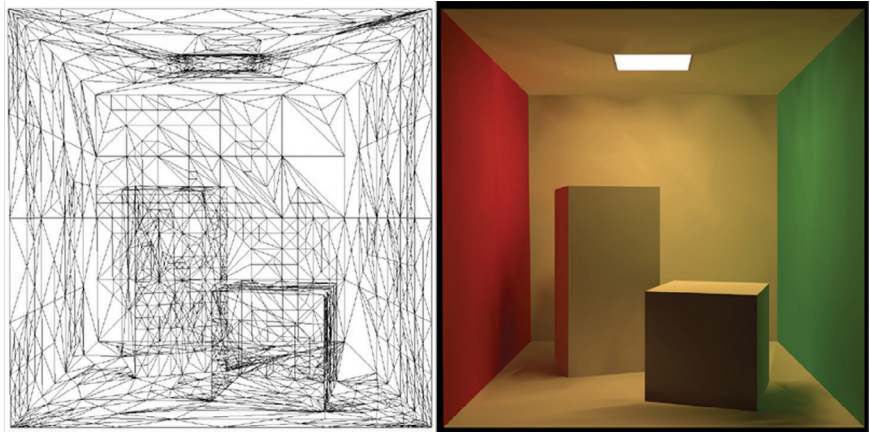


Figure 5-1. Surface meshing (left) in preparation for radiative transfer calculations used to produce a rendering (right).

5.2.2 Ray Tracing. Ray tracing uses the geometric optics model of light transport to perform calculations. Light is modeled as rays, or bundles of rays, moving through an environment, reflecting and scattering off opaque surfaces, refracting and dispersing through transparent or translucent ones.³⁶ The resolution of this method of light transport depends on the number of rays traced. The fidelity of the result depends on how accurately the spatial variance of radiant sources is modeled, and how well radiant or luminous variation across surfaces is modeled. Both depend on the density, and therefore number, of rays. In most software that uses ray tracing, rays are radiant rather than luminous entities. (Refer to ANSI/IES LS-2-20.⁷)

Forward ray tracing initiates rays outward from light sources. These are the 0th-order rays. An example is shown in red in **Figure 5-2**. Each ray, or a bundle of rays, is traced until it encounters a surface that is reflective, transmissive, or both. Depending on the optical properties of the surface, second-order rays are launched from the point of this ray-surface intersection into appropriate directions, and these are each subsequently followed through the environment, producing third-order rays. The number, luminous (or radiant) power, and direction of high-order rays depends on the direction and power of the incident ray, and the reflective, transmissive, refractive, or scattering character of the surface it encounters. This process of high-order ray tracing and generation is repeated until the number of high-order rays involved is small or their luminous or radiant power has been reduced to insignificance. At the end of the ray tracing process, the luminance at a point on a surface in the environment is proportional to the total number and power of rays that are known to leave it.

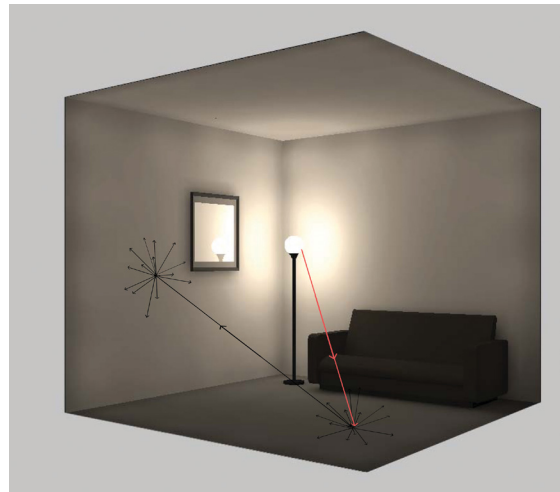


Figure 5-2. An example of forward ray tracing, showing an initial (0th order) ray in red leaving the luminaire and intersecting the floor. Subsequent 1st-order are cast from this intersection point, followed by 2nd-order rays from subsequent points.

Forward ray tracing is seldom used in software for simple quantitative analyses of general lighting systems, but is sometimes used in daylighting analyses to model geometrically or photometrically complex daylight delivery systems. A hybrid combination of forward ray tracing and radiative transfer (see **Section 6.2.5 Mixed Calculations**) is used to determine interreflected surface luminances in the space. It is also used extensively in software for designing and analyzing lighting equipment.³⁷

Backward ray tracing is generally used for generating computer graphic rendered images of lighted environments. In this case, a ray is launched in a particular direction from a fixed point of view, backward toward the environment and through an imaginary image plane. This is designated the 0th-order ray, shown in blue in **Figure 5-3**. The intersection of this ray and the image plane defines an image point. The ray is traced into the environment until it encounters a surface. The luminance of the image point is the luminance of this surface at the ray-surface intersection point. Higher-order rays from this point are, in turn, launched backward into the environment and traced through possibly higher orders of ray-surface interaction until light sources are encountered (the red ray in **Figure 5-3**). From this ray, a backward accounting of the ray's intersections and the surface properties encountered during tracing is made, and the luminance of the image plane at the point defined by the 0th-order ray is incremented appropriately. An accounting of all the high-order rays generated by the 0th-order ray that eventually intersect light sources determine the luminance assigned to the point on the viewing plane. In some implementations of backward ray tracing, when the 0th-order ray encounters a surface, the direct luminance at the intersection point is determined only that one time from each light source.

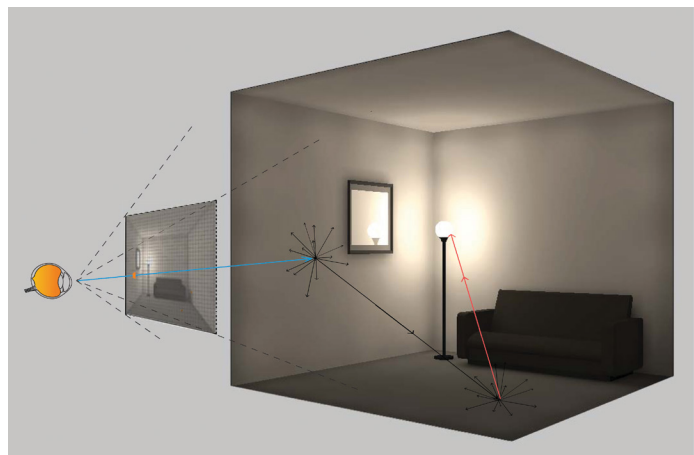


Figure 5-3. An example of backward ray tracing, used to generate an image of a lighted space.

Sophisticated procedures have been developed to guide and optimize the number and tracing direction of rays. Photon mapping involves both forward and backward tracing and is used to generate rendered images of spaces.³⁸ A forward trace is performed first, tracing rays (photons) from sources through the environment. Individual rays are reflected, transmitted, refracted, or absorbed. A ray that encounters a surface or object is absorbed or redirected; statistics and the reflectance or transmittance of the surface or object determines which.

When absorbed, a ray's position and incident direction are recorded, that is, mapped. The mapping is stored and does not depend on rendering viewing point. In a second pass, single-step backward ray tracing is used. A ray is traced from a viewing point through the image plane and into the scene, but only until it intersects a surface. Any first-pass rays that mapped (were absorbed) near this intersection point are used to determine the luminance of the point on the image plane. This procedure does not require first-pass remapping when the point of view for the rendering changes, unlike pure backward ray tracing procedures.

6.0 Renderings Based on Calculations

Surface luminances calculated at dense arrays of points and displayed with the appropriate geometric projection on a high-resolution computer display can produce realistic images of lighted environments that are useful in lighting system design and analysis. Within the limits of geometric, computational, and display details, these images can assume the appearance of photographs and are sometimes described as photorealistic renderings.

These images involve the following types of information and computational work, acquired or performed in the order listed:^{39,40}

1. Geometric description of the environment
2. Definition of surface properties
3. Definition of light source properties
4. Calculation of surface luminances or spectral radiance
5. Transformation of surface photometric or radiometric properties for display
6. Image display

6.1 Overview of Rendering Generation

Renderings of lighted environments are perspective projections of surfaces onto an imagined viewing plane. Which surfaces, or portions of surfaces, are visible depends on the viewing point, the order in which surfaces present themselves to the viewpoint, and the transparency of intervening surfaces. Every visible point on a surface projects to a point on the viewing plane. **Figure 6-1** shows this typical arrangement. The projection points are along lines that converge to the viewing point. If only the corners and edges of the surfaces are displayed, a so-called wireframe view of the space is generated.

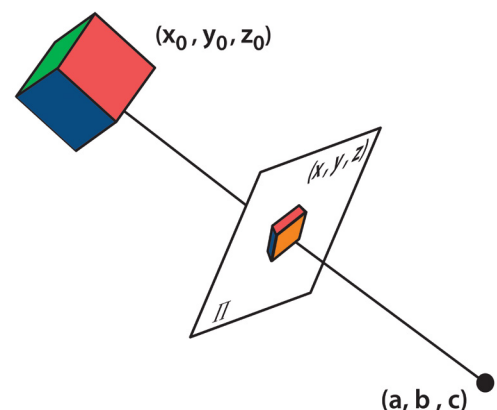


Figure 6-1. Typical perspective rendering projection. The object is projected onto a picture plane located with respect to a viewing position (a, b, c) and perpendicular to the direction of view. All points (x_0, y_0, z_0) in the environment project to points (x, y, z) on the viewing plane.

The photometric or radiometric properties of each point on a visible surface is determined by calculation and become, by projection, the properties of each point on the viewing plane. For most computer systems, these photometric or radiometric properties are converted to three values, each ranging from 0 to 255 that determine the luminance of the red, green, and blue (RGB) elements that comprise a single pixel on the computer display. All or part of the viewing plane is displayed on the computer display, each pixel being set to its calculated RGB value. (See **Section 6.4 Display Properties and Limitations**.)

6.2 Computational Basis

The computational basis for renderings has two principal aspects: whether the calculation is photometric or radiometric, and whether the calculation is based on radiative transfer, ray tracing, or a combination of both.

6.2.1 Photometric Calculations. Some rendering software based only on photometric calculations produce grayscale renderings. Light sources are described with flux or intensity, single values that subsume the spectral properties of the source. Material properties are described with integrated reflectance or transmittance, values that subsume the spectral reflectance or transmittance into a single quantity. Calculations using just these quantities usually employ the so-called “gray assumption”; that is, the spectral properties of sources and surfaces are flat or uniform.

The luminances calculated on this basis can only be mapped to a single perceptual scale: lightness. Photometric calculations have the benefit of requiring only commonly available information about lighting equipment and materials.

6.2.2 Radiometric Calculations. Rendering software based on radiometric calculations can produce color renderings. Sources are characterized by not only a spatial distribution of flux but also a spectral power distribution. Surfaces are described with spectral reflectances and transmittances. The resulting calculations are spectral radiances at points on surfaces. As with any spectral data, these can be transformed into colorimetric quantities and, to the extent permitted by computer displays, can be used to show surface colors. (Refer to *ANSI/IES LS-5-20, Lighting Science: Color*.⁴¹)

In some software, radiant calculations are done using three broad, overlapping bands in the radiometric spectrum, concentrated in the long, medium, and short wavelengths. These are usually referred to as red, green, and blue, or RGB. Surfaces have reflective and transmissive properties specified by three value in these wavelength regions. Unless specific information is available, sources are assumed to have a spectral power distribution of a blackbody radiator of 6500 K. This radiant power is apportioned into the three radiometric bands.

6.2.3 Radiative Transfer Calculations. Surface luminances or radiances can be calculated using the solution of **Equation 5-5** (see **Section 5.2.1 Radiative Transfer**) and surface reflectances. The luminance or radiance of a discretized element is assigned to a point on the element. These points are treated as vertices of triangles that are projected onto the viewing plane and displayed. Points between these vertices have luminances or radiances determined by interpolation.

This final interpolation is usually bilinear and is often done by the computer graphics hardware. In this case, spatial accuracy or resolution depends on the discretization used in the radiative transfer calculation, since that determines the point spacing and position. A mesh with large elements can produce incorrect luminance patterns, especially with luminaires that are close to a surface, as in the case of a wall sconce, a wall slot, or an indirect luminaire.

Rendering systems of this type may exhibit artifacts caused by the failure of triangle edges or vertices to align with geometric edges and intersections, as when surfaces are covered by touching or intersecting surfaces. This can result in areas of inappropriately high or low luminance, called *light-leaks* or *shadow-leaks*. This problem can be solved by

additional geometric constraints on the radiative transfer discretization process, or minimized by calculating the direct component of luminances at a more closely or more carefully placed set of points.

An alternative procedure is to calculate surface luminances at points not necessarily related to the underlying surface discretization, but spaced and located based only on anticipated luminance gradients and surface contact areas or intersections. The luminance at these points is determined not from the underlying surface, but by use of **Equation 5-6** (see **Section 5.2.1 Radiative Transfer**) and approximations of **Equation 3-2** (see **Section 3.2 Illuminance from Area Sources**) to calculate the illuminance at the point and then use the diffuse reflectance of the surface to obtain the luminance. This procedure has the advantage of uncoupling the discretization and subsequent radiative transfer analysis from the calculation desired for displaying surface luminances in renderings. That is, the two can have different spatial resolutions. This procedure has the disadvantage of involving an additional computational step.

Virtually all rendering systems that use radiative transfer are based on diffuse radiative transfer in that reflecting surfaces are assumed diffuse. However, even in the case of a diffuse radiative transfer model, some special types of transfer, such as flux passing through windows, do not have to be assumed diffuse.

6.2.4 Ray Tracing Calculations. Ray tracing calculations are described in **Section 5.2.2 Ray Tracing**. In most cases, backward ray tracing, coupled with the cached results from a forward ray trace, is used to determine the photometric or radiometric properties of points on the viewing plane. In these cases, the position of these points is not determined by forward projection from the environment, but backward projection from the viewing point, through a position exactly corresponding to the position of a pixel on the computer display, and back into the environment. This produces the greatest spatial resolution.

Ray tracing easily accounts for the directional reflectance and transmittance properties of materials. Since individual rays are traced, no assumptions need to be made about their reflected or transmitted distribution. Rather, the bidirectional reflectance and transmittance data describing materials, when available, is used to determine reflected, transmitted, or refracted ray directions.

6.2.5 Mixed Calculations. Some software systems combine radiative transfer and ray tracing. It is possible to approximately account for the visual appearance of specular and semi-specular reflection in an environment by separating the calculation of the direct and interreflected luminance of a surface used in a rendering. One implementation of this procedure calculates the interreflected luminance using the integrated reflectance of the surface, and the direct luminance using the appropriate BRDF. The interreflected component at each point on the surface is calculated using **Equation 5-6** (see **Section 5.2.1 Radiative Transfer**), the direct component by use of **Equation 3-4** (see **Section 3.3 Luminance at a Point**) or its approximation.

Mixed calculations approximate what is rigorously a non-diffuse radiative transfer problem. They are reliable when the amount of non-diffuse transfer in the environment is small—that is, when the number of lumens transported in the environment by non-diffuse reflection is small. This is the case when the specular component of mixed reflectance is small compared to the total reflectance, or when the surface area with non-diffuse reflectances is small compared to the total.

6.3 Adding Realism to Renderings

Texture mapping is a rendering procedure that modifies a surface by the computational equivalent of applying a pattern after all other calculations have been complete.⁴² The pattern can have various amounts of transparency and color, and is often used to simulate wood grain, marble and tile textures, and other surface finishes. Texture mapping can add considerable realism to a rendering. It is important to note, however, that these “textures” are flat; they have the look of the textured surface but without the three-dimensionality; think, “brick” wallpaper.

Bump mapping is a rendering procedure that alters the luminance calculation (often just the direct component) at points on a surface. At each point, a *texture height map* is consulted, and the normal of the surface at that point is altered accordingly. This perturbed normal is used in the calculation of the luminance at the point. Depending on the texture height map, this can give the surface a textured appearance, such as brushed metal, or the skin of an orange.

Other procedures to process surfaces to make them appear more realistic are parallax mapping,⁴³ displacement mapping, normal mapping, and relief mapping.³⁶

6.4 Display Properties and Limitations

Rendering involves the conversion of photometric or radiometric properties of points on a viewing plane to appropriate signals for pixels on a computer display or dots of a color printer. In almost all cases, the gamut of colors and the range of luminances of the array of points on the viewing plane far exceed what can be directly produced by a computer display or, less so, by printer. Color gamuts of common computer displays do not include very saturated colors. (See ANSI/IES LS-5-20.⁴¹) Commonly used computer displays have maximum luminance ratios of approximately 150:1, while real environments often present ratios as high as 75,000:1. Thus, the color gamut and luminance range of the original computation has to be compressed. This compression process is called *tone mapping*.

Tone mapping can also account for a failure to produce deep black, the effect of object size in an image, and the local range of luminance in the field of view. The fundamental task is to produce a stimulus on the computer display or printed page that produces a perception similar to that produced by viewing the actual environment—that is, produce a perceptual metamer for the environment.

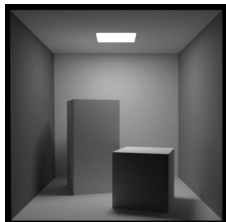
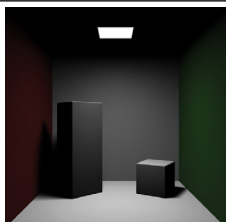
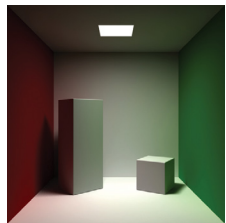
Tone mapping procedures, and the images they produce on commonly available computer displays, have not been extensively evaluated against actual environments, but existing data indicate that rendered images can be useful perceptual metamers.⁴⁴

Tone mapping procedures have been evaluated psychophysically by comparing the images they produce on commonly available displays to images displayed directly on custom display devices with very large color gamuts and luminance ratios⁴⁵ that can accurately reproduce luminance and colors.⁴⁶ Some of these procedures are particularly effective and necessary when compressing images with very high luminance ratios^{47,48} as often occurs when rendering daylight environments.

Automatic tone mapping does not always work without user or observer intervention, and so some software systems permit modifications to the parameters that govern tone mapping to adjust the rendered image. Though useful, and sometimes essential, abuse or incorrect use of these adjustments can produce a realistic or pleasing image, but one that fails to accurately represent the luminous environment.

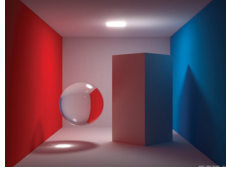
Table 6-1 shows examples of the various rendering processes described above.

Table 6-1. Examples of Rendering Using Different Procedures

Rendering Method	Condition	Rendering
Radiative Transfer with Simple Photometric Meshing	The surfaces were meshed photometrically—that is, discretized into smaller elements where large changes in luminance were anticipated. The radiative transfer solution gave a final exitance to each discretized element. The rendering used the centers of these discretized elements to locate the calculated exitances, and then interpolated smoothly between them. A more elaborate method is to account for the boundaries between discrete elements. In either case, this type of discretization only can produce artifacts. Since the surfaces are not necessarily discretized at points or lines of contact, shading can produce so-called light leaks or shadow leaks. Thus, the lower-left corner of the box in the foreground appears to float.	
Radiative Transfer with Geometric and Photometric Meshing	In this case, the surfaces are discretized not only photometrically but also geometrically, being divided along lines defined by any edge of another surface with which it is in contact. This helps eliminate light leaks and shadow leaks. There are usually more discretized elements.	
Radiative Transfer Color Rendering, Direct Light Only	This rendering was made with direct illumination only; the interreflected component was suppressed. In this case, the spectral reflectivity of the surfaces is taken into account. That is, the “grey assumption” is not made. The left and right walls demonstrate the color (red and green) determined by their spectral reflectance and the SPD of the light source. With no interreflected light, the shadows are mostly umbral and the ceiling is completely unilluminated.	
Ray Tracing Color Rendering, Direct and Interreflected Light	In this case, the interreflected component is accounted for. The deeply saturated colored walls reflect radiation of relatively narrow spectral composition; therefore, the interreflected light on the floor near the left and right walls is faintly red and green, respectively. This so-called “color bleeding” can only be modeled if the radiative transfer is conducted in multiple wavelength bands, with the appropriate radiative power from the luminaire assigned to each band and the appropriate surface reflectances used for each band. The result is a surface exitance for each element in each wavelength band. Alternatively, it has been shown that it is usually a sufficiently accurate approximation to use three broadband analyses, corresponding to the standard red, green, and blue.	

(continued on next page)

Table 6-1. Examples of Rendering Using Different Procedures (continued from previous page)

Rendering Method	Condition	Rendering
Ray Tracing with Diffuse and Perfectly Specular Surfaces	Ray tracing permits the surfaces in the space being modeled to have reflectance characteristics other than perfectly diffuse. In this case, ray tracing accounts for the perfect specularity of the mirror surfaces of the box in the background. In this analysis, the surfaces are either perfectly diffuse or perfectly specular. Though somewhat more elaborate than just a diffuse analysis, the information needed about the mirrors is relatively simple: its specular reflectance. Transfer off the mirror takes place only at pairs of angles obeying Snell's Law.	
Ray Tracing with BRDF Modeling of Surfaces	Ray tracing permits the realistic accounting for surfaces with arbitrary directional reflectances. These are usually characterized by BRDF data. In this case, the sphere in the background has a satin-like finish, with a relatively strong specular component. It should be noted that the bright spot produced on the right, green wall is due to flux from the luminaire reflecting off the sphere. The spectral analyses were done with broadband RGB.	
Photon Mapping and Ray Tracing with Transparency and Refraction	Photon mapping, a kind of view-independent pre-computing process, permits quicker ray tracing and, for a given computation time, more accurate rendering. Focusing effects, such as produced by the transparent sphere and the luminaire, can be modeled.	
Ray Tracing Comparison with Photograph	The image on the left is a color photograph of a box that has been proportioned, papered, and furnished to be like one of the most common, simple tests for computer renderings, the so- called "Cornell box." On the right is a rendering produced with the reflectances and objects of the real box. The rendering is satisfactorily close to the photograph.	

7.0 Evaluating Lighting Analysis Software

Most lighting calculations required in the course of lighting design are performed with computer software specifically written for this purpose. The reliability of the output of lighting software can be determined by sufficiently detailed testing. Even with successful testing, at least an approximate idea of how the software operates, including what approximations it makes and what assumptions it makes, is necessary in order to know the range of problems or situations for which it can be reliably used.

Two ways for testing software used for lighting analysis have been developed: comparison with analytic results and comparison with photometric measurements.

7.1 Analytic Tests

The CIE has established test conditions involving light sources with analytically characterized distributions, surfaces with perfectly diffuse reflectances, and geometry simple enough to permit the analytic determination of, say, illuminances at points. The specification of such conditions can be very precise, and quantities such as illuminance can be calculated directly from well-known theory.⁴⁹ Software can be tested by modeling these analytic conditions and comparing results. Situations have been defined to test various aspects of software. **Figure 7-1** shows an analytic case to test direct illuminance calculations. The luminaire is analytically described, and the illuminance at various points can be determined and compared to software predictions. **Figure 7-2** shows a similar case for testing the direct component of daylighting. More-complicated analytic test cases have been developed for testing interreflected components and the effect of obstructions, as shown in **Figure 7-3**.

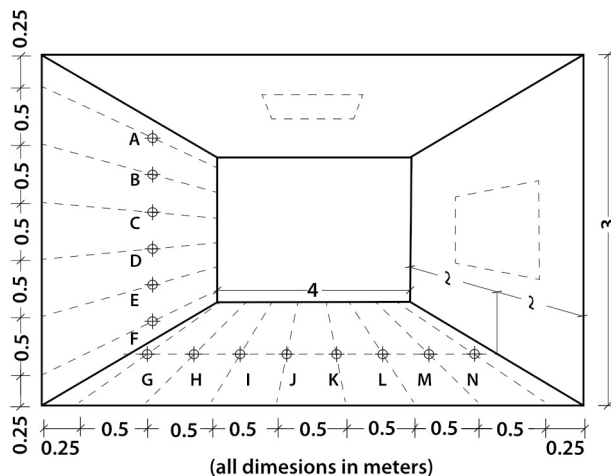


Figure 7-2. CIE computer testing for daylighting: CIE analytic test case for direct-illuminance calculations for daylighting. Points A through N are arranged in a zero reflectance room with either a skylight or window located as shown by the dashed rectangles.

(© Illuminating Engineering Society)

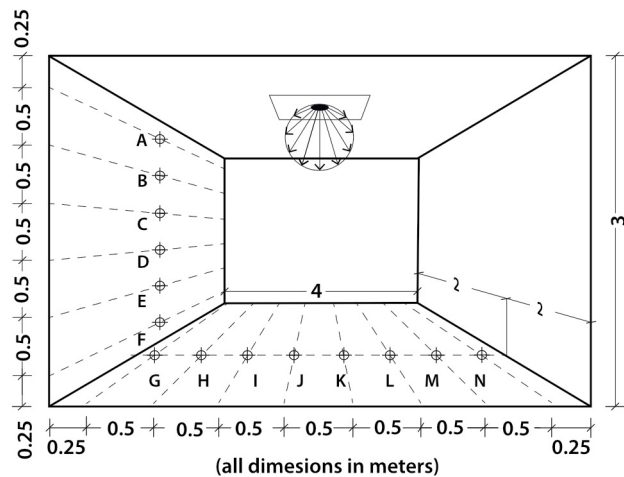


Figure 7-1. CIE computer testing for electric lighting: CIE analytic test case for direct illuminance calculations. Points A through N are arranged in a zero reflectance room beneath a luminaire in the ceiling with an analytic intensity distribution. (© Illuminating Engineering Society)

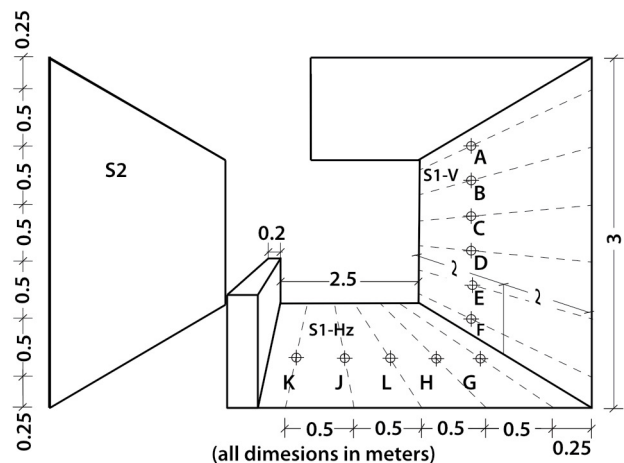


Figure 7-3. CIE Computer Testing for Obstructions: CIE analytic test case for direct illuminance calculations for testing diffuse area source illuminance and the effect of occlusion. (© Illuminating Engineering Society)

7.2 Measurement Tests

The CIE has also developed a set of photometric measurements of illuminance in a room containing various types of luminaires and surface reflectances. The photometric properties of the luminaires, surfaces, measurement equipment, and measurement process were accurately characterized.⁴² **Figure 7-4** shows a plan view of the room.

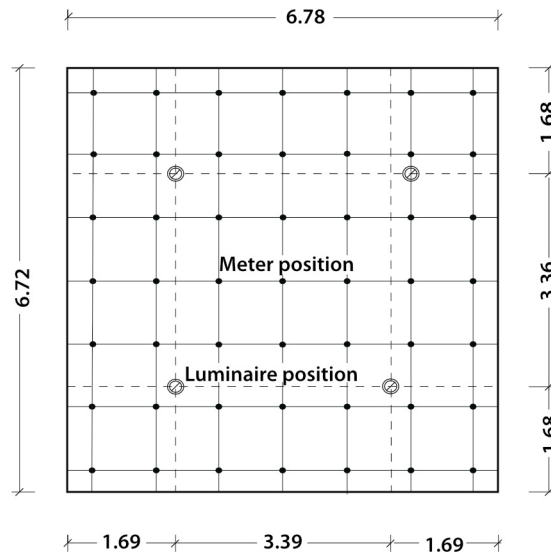


Figure 7-4. CIE Computer testing with measurements: CIE photometric test room, with four luminaires and horizontal illuminance measurement points indicated. Different luminaire distributions and surface reflectances were used.
(© Illuminating Engineering Society)

7.3 Testing Software

The CIE data and other sets of photometric measurements have been used to test commercially available software, with generally good results.⁵⁰⁻⁵⁴

8.0 Factors Affecting Lighting Calculations: Light Loss Factors

Light loss factors (LLF) are used to adjust lighting calculations from laboratory to field conditions. They represent differences in light source lumen output, luminaire output, and surface reflectances between the two conditions. Calculations based on laboratory data alone are likely to provide unrealistic values if not modified by light loss factors. LLFs are assumed to represent independent effects and are therefore multiplicatively cumulative: the total light loss factor is the product of all the applicable factors. No factor should be ignored (i.e., set equal to 1) until investigations justify doing so.

LLFs are divided into recoverable and non-recoverable. *Recoverable factors* are those that can be changed by regular maintenance, such as cleaning and relamping luminaires. *Non-recoverable factors* are those attributed to equipment and site conditions and cannot be changed with normal maintenance.

Additional information may be found in *ANSI/IES RP-36-20, Recommended Practice: Lighting Maintenance*.⁵⁵

9.0 Assessing Computed Results

When lighting calculations are used to assess lighting system performance, it is important that an appropriate method be used. Software is capable of producing a large amount of data, and measures constructed from these data need to be chosen and used carefully if they are to support judgments about lighting system performance.

9.1 Averages

Average usually refers to the mean of several calculated or measured values. The greater the number of values, the more accurate the mean across a given area. Grids of calculation points are usually used, often formed by a rectangular array of rows and columns. Point spacing is determined by the accuracy requirements for the average.

An average can be accurate, but it would never be indicative of the variation in values. For this reason, the average illuminance (or an average of any other quantity) should be used only when the distribution is expected to be relatively uniform across an area. When a localized lighting system is desired, such as task lighting or non-uniform lighting, average illuminance reveals little about the success of the design unless the analysis is limited to the local area. Similarly, an average ceiling luminance produced by an indirect lighting system is not a good measure of performance.

In general, an average value alone is not sufficient to fully describe or evaluate lighting system performance. Information on the uniformity and range of values is also important.

9.2 Minimum and Maximum

If a large number of analysis points are used for calculation, the variability of the lighting can be evaluated, and the minimum and maximum values can be determined and located. The minimums and maximums can be important indicators of the quality of the design, particularly if they deviate significantly from the desired average. In some design situations, maximum and/or minimum design values may be required criteria.

Uniformity is often expressed in terms of a ratio of two quantities. Examples are maximum to minimum, maximum to average, and average to minimum. Different design situations warrant the use of different metrics.

9.3 Coefficient of Variation

Coefficient of variation (CV) is the unit-less ratio of the standard deviation to the mean.

$$CV = \frac{\text{Standard Deviation}}{\text{Mean}} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}}{\bar{x}}, \quad (9-1)$$

where:

$\bar{x}$ = mean of the data values

N = number of data values

Roughly, CV is the average difference from the average, divided or scaled by the average. It expresses how far most of the values are from the average. Put another way, the CV describes the dispersion of the data: the higher the CV, the greater the dispersion and the more different (higher or lower) the values are from the mean. CV is often expressed as a percentage.

9.4 Criterion Ratings

The maximum and minimum values provide little information about the overall distribution of a particular photometric or derived quantity across a space. The criterion rating is a convenient way to obtain greater detail

regarding the distribution of a quantity across a space. The *criterion rating* is the probability that a specific criterion will be met or exceeded anywhere within a defined area. It can be used in addition to (or instead of) concepts such as average, or minimum and maximum levels. Lighting criteria to which this technique may be applied include luminance, illuminance, contrast, visibility metrics, and visual performance metrics.

The criterion rating assumes the name of the criterion being rated. For example, the criterion rating for illuminance is called the illuminance rating. If, for example, an illuminance of 300 lx has been established as the design criterion for a space, then the illuminance rating defines the likelihood that at any point on the work plane the illuminance will be equal to or greater than 300 lx. This criterion rating is determined by evaluating the appropriate quantity at a grid of points covering the area in question.

It is required that the distance between evaluation points not exceed one-fifth the distance from any luminaire to the evaluation plane. The criterion rating is the proportion of points that comply with the criterion, usually expressed as a percentage:

$$\text{Criterion Rating} = \frac{\text{No. of points satisfying criterion}}{\text{No. of points computed or measured}} \quad (9-2)$$

Criterion ratings may be expressed using a notation that lists the rating, in percent, followed by the criterion and separated by the symbol @, which stands for "at." For example, a lighting system producing a luminance of 20 cd/m² over 60% of the specified area may have its luminance criterion rating expressed as 60%@20 cd/m².

10.0 Standardized Calculation Procedures

Some quantities are calculated frequently enough to warrant a standardized calculation procedure to help provide uniform processes, consistent bases for comparisons, and reliable and uniform data. Standardized procedures have been developed for calculating average illuminance and the potential for discomfort glare.

10.1 Calculating Average Illuminance

The lumen method is used in calculating the average illuminance, $\bar{E}$, on a work plane in an interior:

$$\bar{E} = \frac{\text{total flux onto work plane}}{\text{work plane area}}, \quad (10-1)$$

The coefficient of utilization (CU) gives the fraction of lamp lumens that reach the work plane, directly from sources and from interreflections. The CU takes into account the efficiency of the luminaire and the impact of the luminaire distribution and the room surfaces in its derivation. The algorithm for calculating a CU is given in **Annex A – Formulary**. Thus, the number of lumens produced by the lamps and then multiplied by this CU determines the number that reaches the work plane. Thus:

$$E = \frac{(\text{total lamp lumens}) \times \text{CU}}{\text{work plane area}}, \quad (10-2)$$

Since the design objective is usually maintained illuminance, a light loss factor (LLF) is applied to allow for the estimated depreciation in lamp lumens over time, the estimated losses from dirt collection on the luminaire surfaces (including lamps), and other factors that affect luminaire lumen output over time. The formula thus becomes:

$$\bar{E}_{\text{maintained}} = \frac{(\text{total lamp lumens}) \times \text{CU} \times \text{LLF}}{\text{work plane area}}, \quad (10-3)$$

Although design calculations are based on the LLF using both non-recoverable and recoverable factors, it is sometimes necessary to calculate illuminance in a new lighting installation. In such cases, the calculation is performed using the non-recoverable losses, since the recoverable losses will not have occurred at 100 hours, the time at which lamps are nominally at rated lumens.

The lamp lumens in the formula are most conveniently taken as the total rated lamp lumens in the luminaires:

$$E_{\text{maintained}} = \frac{(\text{No. luminaires})(\text{lamps/luminaire})(\text{lumens/lamp}) \times CU \times LLF}{\text{work plane area}} \quad (10-4)$$

If the desired maintained illuminance is known, this equation can be solved for the total number of luminaires needed:

$$\text{No. luminaires} = \frac{E_{\text{maint.}} \times (\text{work plane area})}{(\text{lamps/luminaire})(\text{lumens/lamp}) \times CU \times LLF} \quad (10-5)$$

10.1.1 Determining the CU: The Zonal Cavity Method. In the Zonal Cavity Method, the effects of room proportions, luminaire suspension length, and work plane height upon the coefficient of utilization are respectively represented by the room cavity ratio, ceiling cavity ratio, and floor cavity ratio. These ratios are determined by dividing the room into three cavities, as shown by **Figure 10-1**, and substituting dimensions (in meters or feet) into the following formula:

$$CR = \frac{5H(L + W)}{LW} \text{ or } \frac{2.5H(\text{perimeter length})}{A}, \quad (10-6)$$

where:

CR = cavity ratio

L = cavity length

W = cavity width

H = cavity height, where (see **Figure 10-1**):

H = H_{RC} for the room cavity ratio (RCR)

H = H_{CC} for the ceiling cavity ratio (CCR)

H = H_{FC} for the floor cavity ratio (FCR)

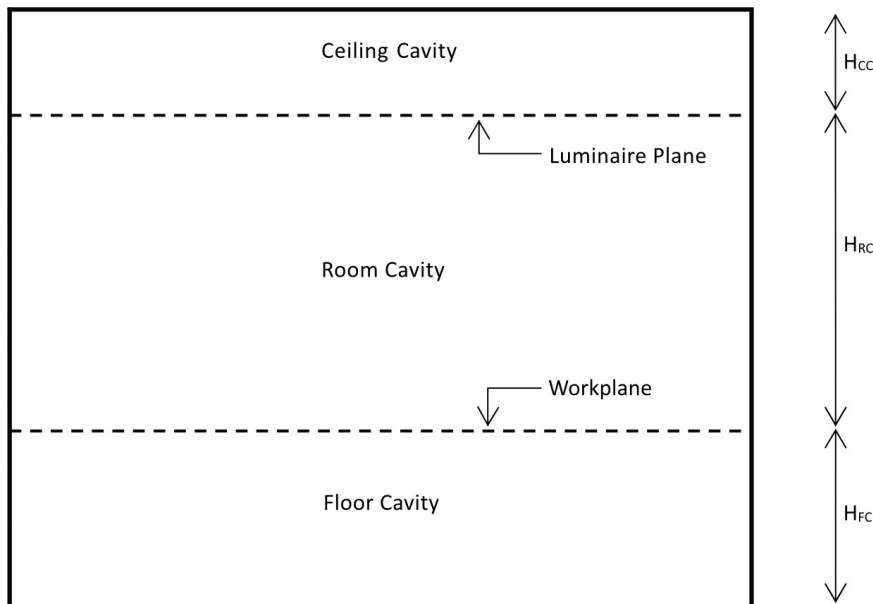


Figure 10-1. Three cavities used in the lumen method: A vertical section through a rectangular room showing heights used in the determination of room, ceiling, and floor cavity ratios for choosing an appropriate CU in the lumen method.

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10.1.2 Effective Cavity Reflectances. A rectangular cavity consists of four walls, each having a reflectance of ρ_w , and a base of reflectance ρ_B (ceiling or floor reflectance). The effective reflectance, ρ_{eff} , of this cavity is the ratio of the flux reflected out to the flux entering the cavity through its opening. If the reflectances are assumed perfectly diffuse and the flux is assumed to enter the cavity in a perfectly diffuse way, it is possible to calculate the effective cavity reflectance using flux transfer theory. The result is:

$$\rho_{\text{eff}} = \frac{\rho_B \rho_w f \left[2 \frac{A_B}{A_w} (1-f) - f \right] + \rho_B f_2 + \rho_w \frac{A_B}{A_w} (1-f)^2}{1 - \rho_B \rho_w \frac{A_B}{A_w} (1-f)^2 - \rho_w \left[1 - 2 \frac{A_B}{A_w} (1-f) \right]}, \quad (10-7)$$

where:

A_B, A_w = areas of the cavity base and walls, respectively

ρ_B, ρ_w = reflectances of the cavity base and walls, respectively

f = form factor between the cavity opening and the cavity base

The form factor needed in **Equation 10-7** can be approximated by:

$$f \approx 0.026 + 0.502 \exp\left(-0.6750 \frac{A_w}{A_B}\right) + 0.470 \exp\left(-0.2975 \frac{A_w}{A_B}\right) \quad (10-8)$$

Equation 10-8 provides a means of converting the combination of wall and ceiling or wall and floor reflectances into a single effective ceiling cavity reflectance, ρ_{CC} , and a single effective floor cavity reflectance, ρ_{FC} . In lumen method calculations, the ceiling, wall and floor reflectances should be initial values. It should be noted that for surface-mounted and recessed luminaires, the CCR equals 0, and the actual ceiling reflectance may be used for ρ_{CC} .

If the reflectance over one of the room surfaces varies considerably, an area-weighted average reflectance should be used—for example, when a high reflectance classroom wall is partially covered by a low reflectance chalkboard.

10.1.3 Limitations. The illuminance computed by the lumen method is an average value that will be representative only if the luminaires are spaced to obtain reasonably uniform illuminance. The calculation of the coefficients of utilization is based on empty interiors having surfaces that exhibit perfectly diffuse reflectance. The average illuminance determined by the lumen method is defined as the total lumens reaching the work plane, divided by the area of the work plane (see **Equation 10-1**). The average value determined this way might vary considerably from that obtained by averaging discrete values of illuminance at several points, especially if the number of points is small or the points do not extend over the entire work plane.

10.2 Calculating Glare

10.2.1 Visual Comfort Probability. Discomfort glare is the sensation of discomfort caused by luminances that are high relative to the average luminance in the field of view. Visual comfort probability (VCP) is the probability that a normal observer does not experience discomfort when viewing a lighting system under defined conditions.

VCP is a metric with limited application. It was developed for lensed fluorescent lighting equipment. It is not valid for use with incandescent HID, or LED equipment. It also cannot be used with luminaires that have an upward component. The procedure has never been proven to accurately model the discomfort caused by parabolic fluorescent luminaires. Parabolic luminaires are much less uniform in luminance than lensed luminaires, and the difference can have a noticeable effect on the comfort of occupants.

Small differences in VCP are not significant. VCP differences of less than 5 points do not indicate a meaningful difference in discomfort glare potential.

The calculation of VCP is covered in **Annex A – Formulary** (in **Section A.8 Calculating Visual Comfort Probability**.)

10.2.2 Unified Glare Rating. The CIE has developed a unified glare rating (UGR) system intended for discomfort glare prediction that has been adopted by many nations. This formula is limited to those situations where the solid angle of the source, ω , is between 0.0003 and 0.1 steradian, inclusive. For example, a troffer luminaire 0.6 m x 1.2 m (2 ft x 4 ft) in a 3-m (10-ft) high ceiling viewed from a distance of 9 m (30 ft) subtends a solid angle of 0.003 steradian. Extensions to this range of applicability have been suggested, but it is not yet clear how accurate they are.⁵⁶ Values of UGR range from 5 to 30, with higher numbers indicating greater discomfort glare.

UGR is calculated from:

$$UGR = 8 \log \left(\frac{0.25}{L_b} \right) \sum_i \frac{L_i^2 \omega_i}{P_i^2}, \quad (10-9)$$

where:

- L_b = luminance (in cd/m²) of the field of view, which does not include luminaire luminance
- L = luminance of a luminaire in the direction of the observer
- ω = solid angle of a luminaire subtended to the observer
- P = position index of luminaire

The position index of a source, P , is an inverse measure of the relative sensitivity to a glare source at different positions throughout the field of view. Selected values or families of curves were published in early references. The formula for calculating P is given by **Equation A-28** in **Annex A – Formulary**.

In most applications, calculation and use of UGR has replaced VCP (see **Section 10.2.1 Visual Comfort Probability**).

Annex A – Formulary

The fundamental equations of illumination engineering are found in **Table A-1**. Methodologies for calculating other important entities are provided in **Sections A.1** through **A.5**, including:

- Configuration factors
- Form factors
- Lumen method coefficients of utilization (CU)
- Spacing criteria (SC)
- Solid angle
- Luminaire efficiency
- Average luminaire luminance

Table A-1. Tabulation of Fundamental Equations

Quality	Condition	Formula	
Illuminance at a point	Point source	$E_p = \frac{I(\theta, \psi) \cos \xi}{D^2}$	
Illuminance at a point	Point source, using an intensity array	$E_p = \frac{I'(\theta, \psi) \cos \xi}{D^2}$	$I'(\theta, \psi)$ interpolated from: $I'(\theta, \psi)$; $I'(\theta_{i+1}, \psi_j)$; $I'(\theta_i, \psi_{j+1})$; $I'(\theta_{i+1}, \psi_{j+1})$ $\theta_i \leq \theta \leq \theta_{i+1}$, and $\psi_j \leq \psi \leq \psi_{j+1}$
Flux on an area	Point source	$\Phi = \int_{A_r} \frac{I(\theta, \psi) \cos \xi}{D^2} dA_r$	
Flux on an area	Point source, using an approximate area integral	$\Phi = \sum_{i=1}^N \frac{I'(\theta, \psi) \cos \xi}{D^2} \Delta A_i$	N = Number of pieces of area ΔA_i = area of i^{th} piece $I'(\theta_i, \psi_i)$ interpolated intensity for i^{th} piece
Illuminance at a point	Area source, arbitrary luminance	$E_p = \int L(\theta, \psi; u, v) \cos \xi d\omega_s$	
Illuminance at a point	Area source, homogeneous luminance	$E_p = \int L(\theta, \psi) \cos \xi d\omega_s$	
Illuminance at a point	Area source, homogeneous luminance, using an approximated-area integral	$E_p = \sum_{i=1}^N \frac{L'(\theta_i, \psi_i) \cos \theta_i \cos \xi_i}{D_i^2} \Delta A_i$	N = Number of pieces of area ΔA_i = area of i^{th} piece $L'(\theta_i, \psi_i)$ = interpolated for i^{th} piece
Illuminance at a point	Area source, using far-field luminous intensity and an approximated-area integral	$E_p = \frac{1}{A} \sum_{i=1}^N \frac{I'(\theta_i, \psi_i) \cos \xi_i}{D_i^2} \Delta A_i$	N = no. of pieces of luminaire ΔA_i = area of i^{th} piece of lum'r $I'(\theta_i, \psi_i)$ = interpolated for i^{th} piece
Illuminance at a point	Area source, homogeneous diffuse exitance	$E_p = M \frac{1}{\pi} \int_{A_s} \frac{\cos \theta \cos \xi}{D^2} dA_s$	M = diff. exitance of area source A_s = entire area of the source
Flux on an area	Area source, arbitrary luminance	$\Phi = \int_{A_r} \int_{\Omega_s} L(\theta, \psi; u, v) \cos \xi d\omega_s dA_r$	Ω_s = entire solid angle of source A_r = entire area of the receiver

(continued on next page)

Table A-1. Tabulation of Fundamental Equations (continued from previous page)

Quality	Condition	Formula	
Flux on an area	Area source, homogeneous luminance, using an approximated-area integral	$\Phi = \sum_{j=1}^K \sum_{i=1}^N \frac{L'(\theta_{ij}, \psi_{ij}) \cos \theta_{ij} \cos \xi_{ij}}{D_{ij}^2} \times \Delta A_i \Delta A_j$	N = no. of source pieces K = no. of receiver pieces $\Delta A_i = i^{\text{th}}$ piece of source $\Delta A_j = j^{\text{th}}$ piece of receiver $L'(\theta_{ij}, \psi_{ij})$ is interpolated for each (i, j)
Flux on an area	Area source, using far-field luminous intensity and an approximated-area integral	$\Phi = \sum_{j=1}^K \frac{1}{A_s} \sum_{i=1}^N \frac{I'(\theta_{ij}, \psi_{ij}) \cos \theta_{ij} \cos \xi_{ij}}{D_{ij}^2} \times \Delta A_i \Delta A_j$	N = no. of source pieces K = no. of receiver pieces $\Delta A_i = i^{\text{th}}$ piece of source $\Delta A_j = j^{\text{th}}$ piece of receiver $I'(\theta_{ij}, \psi_{ij})$ is interpolated for each (i, j) A_s = entire area of the source
Flux on an area	Area source, homogeneous diffuse exitance	$\Phi_{s \rightarrow r} = \frac{M_s}{\pi} \int_{A_r} \int_{\Omega_s} \cos \xi d\omega_s dA_r = \frac{M_s}{\pi} \int_{A_r} \int_{A_s} \frac{\cos \theta \cos \xi}{D^2} dA_s dA_r$	
Configuration factor		$c = \frac{1}{\pi} \int_{A_s} \frac{\cos \theta \cos \xi}{D^2} dA_s$	
Form factor		$f_{s \rightarrow r} = \frac{1}{\pi A_s} \int_{A_r} \int_{A_s} \frac{\cos \theta \cos \xi}{D^2} dA_s dA_r$	
Illuminance at a point	Uniform diffuse source	$E_p = M_s c$	
Average illuminance on a surface	Uniform diffuse source	$\bar{E}_r = \bar{M}_s f_{r \rightarrow s}$	

A.1 Calculating Configuration Factors

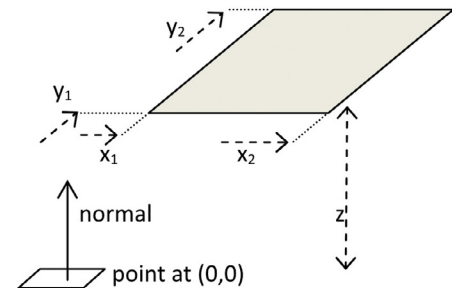
Equation 3-9 (see **Section 3.6 Diffuse Surfaces**) gives the general form for calculating a radiative transfer configuration factor. For many geometric settings, a convenient coordinate system can be found and each component of the integral made specific to that coordinate system. In many cases, the resulting double integral (i.e., area integral) has a closed form. Extensive tabulations of equations for specific geometries are available.⁵⁷ Equations are given here for commonly occurring geometric settings found in architectural lighting.

Point in a Plane and a Rectangle in a Parallel Plane (see Figure A-1):

$$c = \frac{1}{2\pi} \sum_{i=1}^2 \sum_{j=1}^2 F(x_i, y_j) (-1)^{i+j}, \quad (\text{A-1})$$

where:

$$F(x_i, y_j) = \frac{x_i}{\sqrt{x_i^2 + z^2}} \tan^{-1} \left(\frac{y_j}{\sqrt{x_i^2 + z^2}} \right) + \frac{y_j}{\sqrt{y_j^2 + z^2}} \tan^{-1} \left(\frac{x_i}{\sqrt{y_j^2 + z^2}} \right)$$

**Figure A-1. Geometry for Equation A.1.**

Point in a Plane and a Rectangle in a Perpendicular Plane
(see Figure A-2):

$$c = \frac{x}{2\pi} \sum_{i=1}^2 \sum_{j=1}^2 F(y_i, z_j) (-1)^{i+j}, \quad (\text{A-2})$$

where:

$$F(y_i, z_j) = \frac{-1}{\sqrt{x^2 + z_j^2}} \tan^{-1} \left(\frac{y_j}{\sqrt{x^2 + z_j^2}} \right)$$

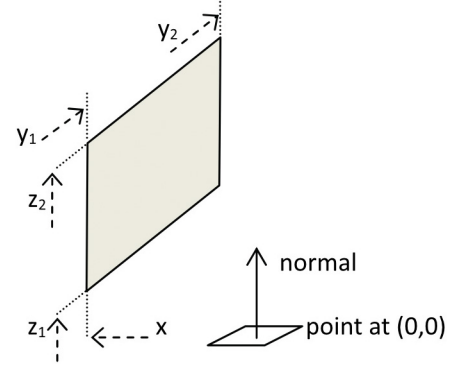


Figure A-2. Geometry for Equation A.2.

Point in a Plane and a Circle in a Parallel Plane (see Figure A-3a):

$$c = \frac{1}{2} \left(1 - \frac{1 + H^2 - R^2}{\sqrt{Z^2 - 4R^2}} \right), \quad (\text{A-3a})$$

where:

$$H = \frac{h}{a};$$

$$R = \frac{r}{a};$$

$$Z = 1 + H^2 + R^2$$

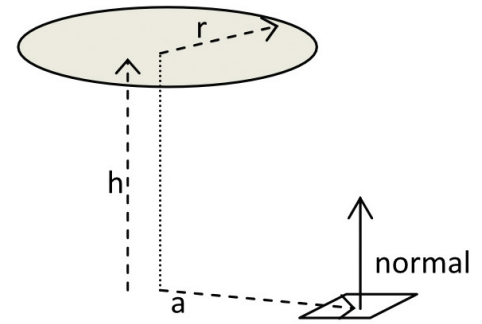


Figure A-3a. Geometry for Equation A-3a.

Point in a Plane and Circle in a Parallel Plane, with the normal to the element passing through the center of the circle
(see Figure A-3b):

$$c = \frac{r^2}{h^2 + r^2}, \quad (\text{A-3b})$$

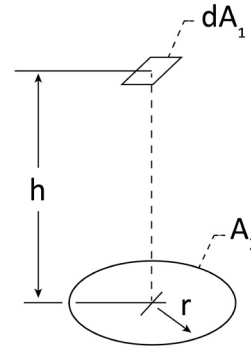


Figure A-3b. Geometry for Equation A-3b.

Point in a Plane and a Circle in a Perpendicular Plane
(see Figure A-4):

$$c = \frac{H}{2} \left(\frac{Z}{\sqrt{Z^2 - 4R^2}} - 1 \right), \quad (\text{A-4})$$

where:

$$H = \frac{h}{l};$$

$$R = \frac{r}{l};$$

$$Z = 1 + H^2 + R^2$$

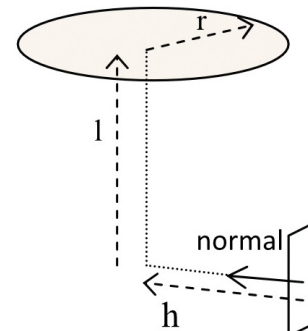


Figure A-4. Geometry for Equation A.4.

A.2 Calculating Form Factors

Equation 3-13 (see **Section 3.6 Diffuse Surfaces**) gives the general form for calculating radiative transfer form factors. As with configuration factors, some geometric settings permit closed-form integration. Equations are given here for commonly occurring geometric settings found in architectural lighting. These equations assume that there are no occluding elements between the surfaces involved; i.e., any point on the first surface has a complete view of the second surface.

Rectangles in Parallel Planes with Parallel or Perpendicular Edges

(see **Figure A-5**):

$$F_{1 \rightarrow 2} = \frac{z^2}{2\pi A_1} \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^2 \sum_{m=1}^2 H(u_i, v_j, x_k, y_m) (-1)^{i+j+k+m}, \quad (\text{A-5})$$

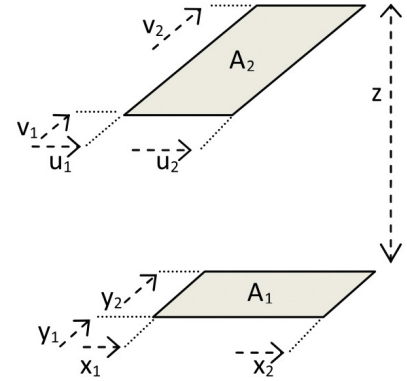


Figure A-5. Geometry for Equation A.5.

where:

$$H(u_i, v_j, x_k, y_m) = b \sqrt{1+a^2} \tan^{-1} \left(\frac{b}{\sqrt{1+a^2}} \right) + a \sqrt{1+b^2} \tan^{-1} \left(\frac{a}{\sqrt{1+b^2}} \right) - \frac{1}{2} \ln(1+a^2+b^2)$$

$$a = \frac{x_k - u_i}{z};$$

$$b = \frac{y_m - v_j}{z}$$

Rectangles in Perpendicular Planes with Parallel or Perpendicular Edges (see **Figure A-6**):

$$F_{1 \rightarrow 2} = \frac{1}{2\pi A_1} \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^2 \sum_{m=1}^2 G(v_i, u_j, x_k, y_m) (-1)^{i+j+k+m}, \quad (\text{A-6})$$

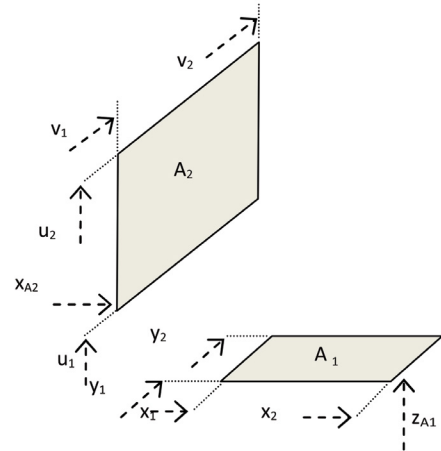


Figure A-6. Geometry for Equation F-6.

where:

$$G(v_i, u_j, x_k, y_m) = a \sqrt{c^2 + b^2} \tan^{-1} \left(\frac{a}{\sqrt{c^2 + b^2}} \right) + \frac{1}{4} (a^2 - b^2 - c^2) \ln(a^2 + b^2 + c^2);$$

$$a = y_m - v_i;$$

$$b = zA_1 - u_j;$$

$$c = x_k - xA_2$$

A.3 Calculating Lumen Method Coefficients of Utilization

Tables of coefficients of utilization (CUs) can be prepared by systematic procedures. It is desirable to standardize the process for producing published tables of these values in order to prevent misunderstandings and to facilitate direct comparisons of the data for different luminaires. These coefficients are derived from the equations described in

Section 5.2.1 Radiative Transfer. The basic assumptions used to develop the zonal cavity coefficients are:

- Room surfaces are Lambertian reflectors
- The incident flux on each surface is uniformly distributed over that surface
- The luminaires are uniformly distributed throughout the room (uniformly dense but not necessarily in a uniform pattern)
- The room is empty
- The room surfaces are spectrally neutral

Full tables of CUs provide values for at least these ceiling reflectances: 0.80, 0.70, 0.50, 0.30, 0.10, and 0.0. At each of these ceiling reflectances, values are provided for at least these wall reflectances: 0.70, 0.50, 0.30, and 0.0. It is recognized that space limitations often necessitate abridgements. In that case, only the columns for $\rho_{CC} = 80\%$, 50%, and 10% are recommended for luminaires having 0% to 35% of their output in the 90° to 180° zone; and 80%, 70%, and 50% for luminaires having over 35% of their output in that zone. Also, the $\rho_{CC} = 10\%$ columns are not required for abridged tables. It is recommended that CUs be published to two decimal places.

The computation of CUs is performed according to the following algorithm:

1. Define 18 conic solid angle zones of 10° width from the nadir to the zenith about the luminaire, as shown in **Figure A-7**, where the index of each zone, N , is an integer between 1 and 18, inclusive.

2. Determine the flux Φ_N (lumens) in the various zones:

- The flux in a conic solid angle (see **Figure A-8**) is given by:

$$\Phi_N = 2\pi I_{\theta_N} (\cos\theta_N - \cos\theta_{N+1}), \quad (\text{A-7})$$

where:

I_{θ_N} = mid-zone intensity, in cd, for the N^{th} zone

θ_N, θ_{N+1} = bounding cone angles

- If the intensity is not rotationally symmetric about the vertical axis, average the intensity about the vertical axis at each vertical angle θ . It is critical that the intensity be sampled at equal angular intervals about the vertical axis. For example, if the intensity is known for three vertical planes [$I_{\theta,90^\circ}$ (perpendicular), $I_{\theta,45^\circ}$ and $I_{\theta,0^\circ}$ (parallel)], then:

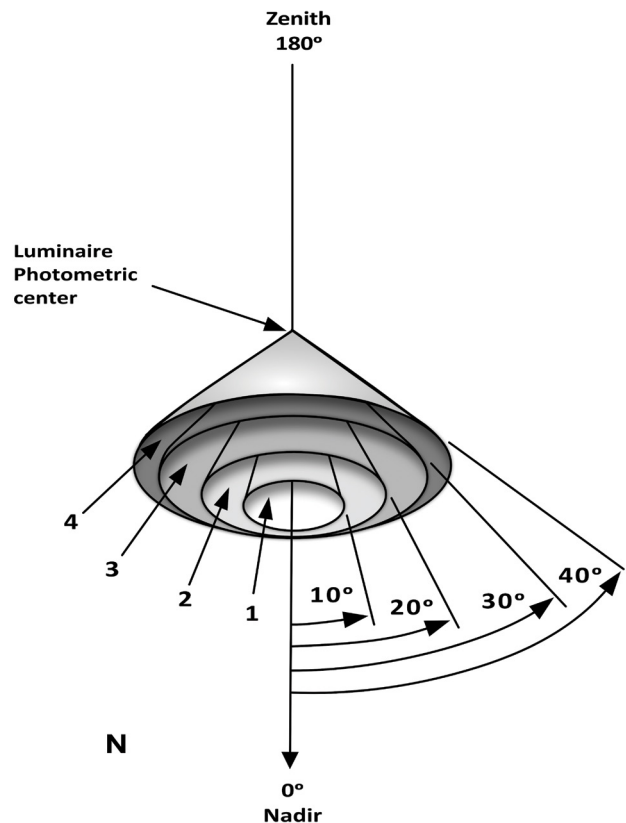


Figure A-7. Conic solid angle zones of 10° width for use in calculating zonal flux. (© Illuminating Engineering Society)

$$I_{\theta} = \frac{1}{4}(I_{\theta,0^{\circ}} + 2I_{\theta,45^{\circ}} + I_{\theta,90^{\circ}}), \quad (\text{A-8})$$

- While three planes are sufficient for luminaires of nominal rotational symmetry, photometric data at 15° or 22.5° increments about the vertical axis are preferred for luminaires without this symmetry.
- If the intensity is taken at 10° vertical intervals ($\theta = 5^{\circ}, 15^{\circ}, 25^{\circ}, \dots$), then the flux Φ_N is determined by the application of **Equation A-7** to the full zone. It is preferred to have intensity values at 5° vertical angles ($\theta = 2.5^{\circ}, 7.5^{\circ}, 12.5^{\circ}, \dots$). Zone N is then divided into two parts, **Equation A-7** is applied to each part, and the resulting flux is summed.

3. Determine the additional flux functions:

$$\Phi_{\text{luminaire}} = \sum_{N=1}^{18} \Phi_N \quad (\text{A-9})$$

$$\eta_{\text{down}} = \frac{1}{\Phi_{\text{lamps}}} \sum_{N=1}^9 \Phi_N \quad (\text{A-10})$$

$$\eta_{\text{up}} = \frac{1}{\Phi_{\text{lamps}}} \sum_{N=10}^{18} \Phi_N \quad (\text{A-11})$$

where:

$\Phi_{\text{luminaire}}$ = total flux emitted by the luminaire

Φ_{lamps} = total flux emitted by the lamps in the luminaire

η_{down} = proportion of lamp flux leaving the luminaire in a downward direction

η_{up} = proportion of lamp flux leaving the luminaire in an upward direction

4. Determine the direct ratio, D_{RCR} , related to the fraction of luminaire flux below the horizontal that is directly incident on the work plane:

$$D_{RCR} = \frac{1}{\eta_{\text{down}} \Phi_{\text{lamps}}} \sum_{N=1}^9 K_{RCR,N} \Phi_N, \quad (\text{A-12})$$

where:

RCR = room cavity ratio; between 1 and 10, inclusive

$K_{RCR,N}$ = zonal multipliers

The zonal multiplier is the fraction of downward-directed flux directly incident on the work plane (lower surface of room cavity) for each zone N . The zonal multipliers are functions of the RCR:

$$K_{RCR,N} = \exp(-A \cdot RCR^B), \quad (\text{A-13})$$

where:

A and B are constants and are given in **Table A-2**.

5. Determine parameters C_1 , C_2 , C_3 , and C_0 as an intermediate step. In the formulas below, ρ_w is the wall reflectance, ρ_{cc} is the ceiling cavity reflectance, and ρ_{fc} is the floor cavity reflectance, which is taken as 0.2 for standard CU tables. $F_{cc \rightarrow fc}$ is the form factor from the ceiling cavity to the

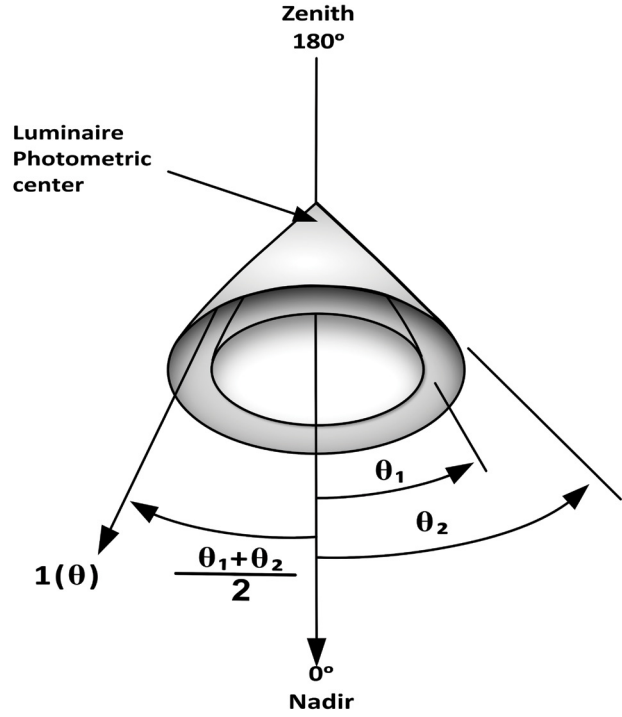


Figure A-8. Angles used in calculating zonal flux.

(© Illuminating Engineering Society)

Table A-2. Constants for Zonal Multiplier Equation

Zone (N)	A	B
1	0	0
2	0.041	0.98
3	0.070	1.05
4	0.100	1.12
5	0.136	1.16
6	0.190	1.25
7	0.315	1.25
8	0.640	1.25
9	2.10	0.80

floor cavity, shown in **Table A-3**. For computer software, **Equation A-5** (see **Section A.2 Calculating Form Factors**) can be used to determine $F_{CC \rightarrow FC}$.

$$C_1 = \frac{(1 - \rho_w)(1 - f_{CC \rightarrow FC}^2)RCR}{2.5\rho_w(1 - f_{CC \rightarrow FC}^2) + RCR \cdot f_{CC \rightarrow FC}(1 - \rho_w)} \quad (\text{A-14})$$

$$C_2 = \frac{(1 - \rho_{CC})(1 + f_{CC \rightarrow FC})}{1 + \rho_{CC}f_{CC \rightarrow FC}} \quad (\text{A-15})$$

$$C_3 = \frac{(1 - \rho_{FC})(1 + f_{CC \rightarrow FC})}{1 + \rho_{FC}f_{CC \rightarrow FC}} \quad (\text{A-16})$$

$$C_0 = C_1 + C_2 + C_3 \quad (\text{A-17})$$

Table A-3. Form Factors for RCRs

RCR	Form Factor
0	1.000
1	0.270
2	0.689
3	0.579
4	0.489
5	0.415
6	0.355
7	0.306
8	0.265
9	0.231
10	0.202

6. Determine the CU for each combination of reflectances and RCR:

$$CU_{RCR; \rho_{CC}; \rho_w; \rho_{FC}} = \frac{2.5\rho_w C_1 C_3 (1 - D_{RCR}) \eta_{down}}{RCR(1 - \rho_w)(1 - \rho_{FC}) C_0} + \frac{\rho_{CC} C_2 C_3 \eta_{up}}{(1 - \rho_{CC})(1 - \rho_{FC}) C_0} + \left[1 - \frac{\rho_{FC} C_3 (C_1 + C_2)}{(1 - \rho_{FC}) C_0} \right] \left(\frac{D_{RCR} \eta_{down}}{1 - \rho_{FC}} \right), \quad (\text{A-18})$$

7. The above equations can be used to calculate the CU when the RCR equals zero, but the forms of the equations need to be arranged to avoid division by zero. It is simplest to use the following relationships:

$$CU_{0; \rho_{CC}; \rho_w; \rho_{FC}} = \frac{\eta_{down} + \rho_{CC} \eta_{up}}{1 - \rho_{CC} \rho_{FC}}, \quad (\text{A-19})$$

A.4 Calculating Luminaire Spacing Criteria

The luminaire spacing criterion⁵⁸ (SC) is a classification technique for interior luminaires relating to the spread or distribution of the illuminance on a horizontal plane. It tests the uniformity of horizontal illuminance at two pairs of selected points to estimate the probable extreme limit of acceptable luminaire spacing. It is not a specification of the spacing-to-mounting-height ratio to be used in a lighting installation, and in fact, installation of luminaires at this nominal value may produce a poor lighting system.

The purpose of this classification technique is to aid designers in rapidly assessing one aspect of the potential of a luminaire with respect to its applications. It gives some idea about the distribution of flux from a luminaire and its subsequent effect on lighting system parameters, using only a single number. The basis of the luminaire spacing criterion is the horizontal illuminance on the work plane due to direct illuminance from nearby luminaires. As a first approximation, it is assumed that this represents a limiting case, since the reflected component of illuminance and the illuminance due to more distant luminaires generally increases the uniformity of horizontal illuminance from point to point.

The calculation of luminaire spacing criteria (SC) is based on the following two premises:

1. When two similar conventional luminaires are near their maximum spacing, the illuminance directly under a luminaire (P) is principally due to the overhead luminaire (A) (see **Figure A-9a**). Further, a very probable point of low illuminance will be at the midpoint between two luminaires (Q). The maximum spacing at a given mounting height above the work plane is chosen such that the illuminance halfway between the two luminaires (Q) and due to both luminaires (A and B) equals the illuminance under one (P) due to that one luminaire (A) only.
2. Another likely point (R) for low illuminance is at the center of a square array of adjacent luminaires (see **Figure A-9b**). The maximum spacing at a given mounting height above the work plane is chosen such that the illuminance at the center of the array of luminaires (R) due to all four luminaires (A, B, C, D) equals the illuminance under one due to that one luminaire (A) only.

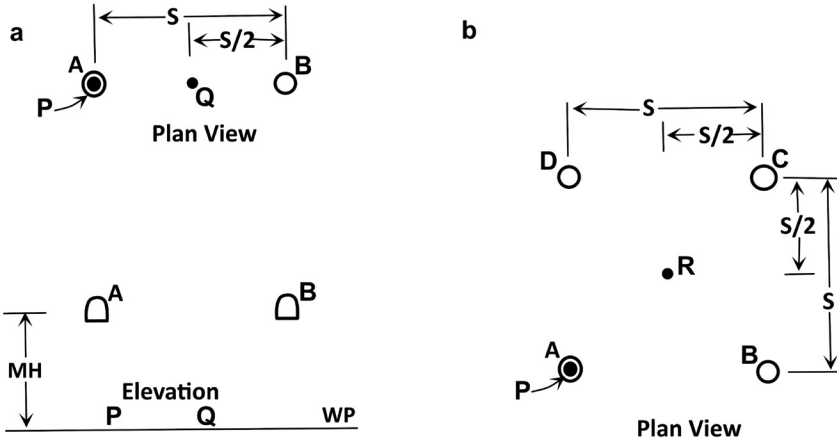


Figure A-9. Points Q and R represent points of most probably low illuminance for (a) a configuration of 2 luminaires, and (b) a square array of 4 luminaires.

The maximum spacing (expressed as a spacing-to-mounting-height ratio that is unit-less) that fulfills each of the above conditions can be determined using the intensity distribution of the luminaire. For the purpose of establishing this criterion, it is assumed that the inverse-square law is valid. This is the only assumption for the computations.

For luminaires with azimuthally symmetric intensity distributions, a single azimuthal plane of data can be used. Otherwise, it is necessary to use three planes of data—usually $\psi = 0^\circ$, $\psi = 45^\circ$, and $\psi = 90^\circ$ —and report two values of SC, one associated with $\psi = 0^\circ$, (“parallel”) and one associated with $\psi = 90^\circ$, (“perpendicular”).

For premise 1, above, assuming an arbitrary height, h , above the work plane:

$$\frac{E_{\text{below}}}{h^2} = \frac{E_{\text{between}}}{h^2} = \frac{2I(\theta, \psi) \cos^3 \theta}{h^2}, \quad (\text{A-20})$$

thus:

$$I(0, \psi) = 2I(\theta, \psi) \cos^3 \theta,$$

where:

E_{below} = illuminance directly beneath one luminaire due to that luminaire only

E_{between} = illuminance midway between luminaires, due to both luminaires

ψ = angle of azimuthal plane used to determine the criterion

θ = the elevation angle

The elevation angle θ is found that gives intensities that satisfy **Equation A-20**:

$$I(\theta, \psi) = \frac{I(0, \psi)}{2 \cos^3 \theta} \quad (\text{A-210})$$

Finding the value of θ that satisfies **Equation A-21** usually requires search and interpolation. The spacing criterion for two luminaires, SC_2 , is then:

$$SC_2 = \frac{h \tan \Theta_2}{h} = \tan \Theta_2, \quad (\text{A-22})$$

where:

Θ_2 = angle θ satisfying **Equation A-21**.

This procedure is repeated considering four luminaires (premise 2, above). In this case:

$$I(\theta, \psi) = \frac{I(0, \psi)}{4 \cos^3 \theta}, \quad (\text{A-23})$$

where:

$$\psi = 45^\circ$$

Next, finding the value Θ_4 that satisfies **Equation A-23** gives:

$$SC_4 = \frac{h \tan \Theta_4}{h} = \tan \Theta_4 \quad (\text{A-24})$$

The spacing criterion for the luminaire is the smaller of SC_2 and SC_4 . However, a photometric report will sometimes include both.

The luminaire spacing criterion only suggests a maximum spacing at which the horizontal illuminance is reasonably uniform. When other criteria are considered, such as overlap between luminaires, vertical illuminance, shadowing, and illuminance distribution above the work plane, it generally is found that luminaires need to be installed at some spacing-to-mounting height ratio less than the value of the luminaire spacing criterion.

A.5 Calculating Solid Angle

The relevant geometry is shown in **Figure A-10**.

$$\text{Solid angle (sr)} = (x^2 \cos \theta) / d^2, \quad (\text{A-25})$$

and

$$\text{Visual angle} = \tan^{-1}(x \cos \theta) / d, \quad (\text{A-26})$$

where:

x = the dimension (length or width) of the critical detail of the target

θ = the viewing angle

d = the viewing distance (x and d are in the same units)

Note: Only one dimension, x , is defined for the critical detail of the target. Visual performance for two different targets subtending the same area will be the same, even if the targets have different aspect ratios; e.g., a square-shaped target versus a long, rectangular-shaped object.^{59,60}

How to interpret:

Small size: 4.0×10^{-6} sr or smaller (solid angle), but not near the acuity limit*

Large size: Larger than 4.0×10^{-5} sr

Notes:

- $1^\circ = 0.0175$ radians = 60 arcmin
- 1 sr = 66° visual angle for a circular target
- For a cone where θ is the half-cone angle, solid angle = $2\pi (1 - \cos \theta)$

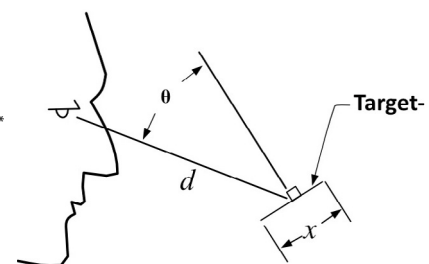


Figure A-10. Determination of visual task parameters. © Illuminating Engineering Society

This division, like that of contrast, is based on the plateau-escarpment nature of visual performance.^{1,2}

Figure A-11 shows numerous examples of visual angle and solid angle in real-world applications.

* It should be noted that contrast threshold and the acuity limit are dependent on background luminance, duration of presentation, color, surround conditions, and in general, any number of factors that affect visibility, including those idiosyncratic to the viewer. Above a contrast of 0.3 and a size of 4.0×10^{-6} sr, these factors are not very important to visual performance.

Printed reading task from 15 in. (40 cm)

Typeface size	Visual angle (°)*	Solid angle (sr) [†]
6 point	0.03	1.7×10^{-6}
8 point	0.04	3.1×10^{-6}
10 point	0.05	4.8×10^{-6}
12 point	0.06	6.9×10^{-6}
14 point	0.07	9.4×10^{-6}
24 point	0.12	2.8×10^{-5}
36 point	0.18	6.2×10^{-5}

*Angular width of single character stroke (vertical stroke, Times typeface).

[†]Average solid angle of total printed area of character for numerical digits (see reference 1).

Viewing a square-shaped object from 100 ft (30 m)

Object size	Visual angle (°)	Solid angle (sr)
3 x 3 in. (7.5 x 7.5 cm)	0.14	6.3×10^{-5}
6 x 6 in. (15 x 15 cm)	0.29	2.5×10^{-5}
12 x 12 in. (30 x 30 cm)	0.57	1.0×10^{-4}

Wire sizes (diameter in cross section) viewed from 15 in. (40 cm)

Wire size	Visual angle (°)	Solid angle (sr)
American Wire Gauge (AWG) 30 (0.25 mm diameter)	0.04	3.9×10^{-7}
AWG 24 (0.51 mm diameter)	0.07	1.6×10^{-6}
AWG 20 (0.81 mm diameter)	0.12	4.1×10^{-6}
AWG 16 (1.29 mm diameter)	0.18	1.0×10^{-5}
AWG 12 (2.05 mm diameter)	0.29	3.3×10^{-5}
AWG 8 (3.28 mm diameter)	0.47	6.7×10^{-5}

Circular drilled holes viewed from 15 in. (40 cm)

Hole diameter	Visual angle (°)	Solid angle (sr)
0.01 in. (0.25 mm)	0.04	3.5×10^{-7}
0.02 in. (0.51 mm)	0.07	1.4×10^{-6}
0.03 in. (0.76 mm)	0.11	3.1×10^{-6}
0.04 in. (1.02 mm)	0.15	5.6×10^{-6}

Figure A-11. Examples of common visual angles and solid angles.

A.6 Calculating Luminaire Efficiency

The luminaire efficiency is simply the ratio of the lumens leaving the luminaire to the total lumens produced by the lamps:

$$\text{Luminaire efficiency} = \Phi_{\text{luminaire}} / \Phi_{\text{lamps}} \quad (\text{A-27})$$

The luminaire efficiency is considered in the determination of the CU.

A.7 Calculating Average Luminaire Luminance

The average luminaire luminance is a means of evaluating the effect that a luminaire has on visual comfort, as well as veiling reflections on task surfaces such as VDTs. A value can be easily determined in any direction, given the photometric characteristics of the luminaire. Since luminance is expressed in candelas per unit area, the average luminance in a direction (θ, ψ) , specified in spherical coordinates, is:

$$L(\theta, \psi)_{\text{avg}} = \frac{I(\theta, \psi)}{A_{\text{luminaire}} \cos \theta} , \quad (\text{A-28})$$

where:

$I(\theta, \psi)$ = intensity of the luminaire in the direction (θ, ψ)

$A_{\text{luminaire}}$ = surface area of the luminous element(s) of the luminaire

A.8 Calculating Visual Comfort Probability (VCP)

Equations for the calculation of the VCP are derived from correlating photometric and geometric characteristics of simple lighting patterns with discomfort glare assessments of observers.⁶¹⁻⁶⁶ Experiments in simulated rooms have been used to confirm the extension from the laboratory to actual lighting installations.^{67,68} This system was tested and validated using lensed direct fluorescent systems only. VCP should not be applied to very small sources such as incandescent, high-intensity discharge, or LED luminaires, to very large sources such as ceiling and indirect systems, or to non-uniform sources such as parabolic reflectors.

To calculate the VCP,⁶⁹⁻⁷³ several intermediate calculations first need to be performed. The position index, P , of a source is an inverse measure of the relative sensitivity to a glare source at different positions throughout the field of view. Selected values or families of curves were published in early references. P is given by⁷⁴:

$$P = \exp \left[(35.2 - 0.31889\alpha - 1.22 e^{-2\alpha/9}) 10^{-3} \beta + (21 + 0.26667\alpha - 0.002963 \alpha^2) 10^{-5} \beta^2 \right] , \quad (\text{A-29})$$

where:

α = angle from vertical of the plane containing the source and the line of sight (see **Figure A-12**), in degrees

β = angle between the line of sight and the line from the observer to the source

The average luminance for the entire field of view, F_v , is approximated by:

$$F_v = (L_w \omega_w + L_f \omega_f + L_c \omega_c + \sum L_s \omega_s) / 5 , \quad (\text{A-30})$$

where:

L_w = average luminance of the walls, in cd/m^2

L_f = average luminance of the floor, in cd/m^2

L_c = average luminance of the ceiling, in cd/m^2

L_s = average luminance of the source, in cd/m^2

ω_w = solid angle subtended at the observer by the walls, in sr

ω_f = solid angle subtended at the observer by the floor, in sr

ω_c = solid angle subtended at the observer by the ceiling, in sr

ω_s = solid angle subtended at the observer by the source, in sr

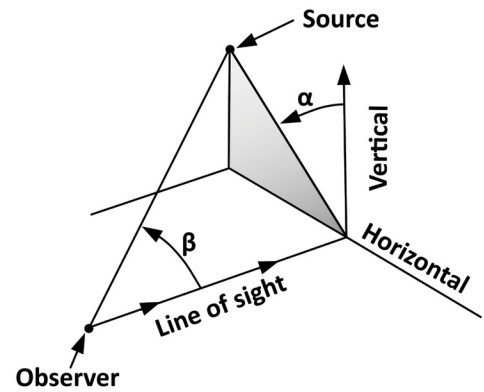


Figure A-12. Geometry defining position index P as used in VCP calculations.

The 5 in the denominator arises from the assumption that the total field of view is 5 sr.⁷⁴

A function Q has been developed for use in the calculation of the VCP. This function is given by:

$$Q = 20.4\omega_s + 1.52\omega_s^{0.2} - 0.075 , \quad (\text{A-31})$$

where ω_s is the solid angle subtended at the observer by the source, in steradians (sr).

The values of P , F_v , and Q are used to determine the index of sensation, M :

$$M = 0.50 L_s Q / P F_v^{0.44} \quad (\text{A-32})$$

The luminance in the above equation is expressed in cd/m^2 . The factor 0.50 in the numerator allows for the use of these units.

From the index of sensation of each source, a discomfort glare rating (DGR) can be calculated for the full field of view. The DGR is a metric of discomfort that increases as discomfort increases; it is also used in calculating the VCP. It is given by:

$$DGR = \left(\sum_{i=1}^n M_i \right) n^{-0.0914} \quad (\text{A-33})$$

where:

n = number of sources in the field of view

M_i = index of sensation for the i^{th} source

The relation between DGR and VCP may be found from a graph such as **Figure A-13** or may be calculated using:

$$VCP = \frac{100}{\sqrt{2\pi}} \int_{-\infty}^{6.374 - 1.3227 \ln DGR} \exp(-t^2/2) dt \quad (\text{A-34})$$

These basic relations may be applied through a variety of techniques. Generally, the room cavity concept is used, in which the actual ceiling and floor luminances and solid angles are replaced by their equivalent cavity values. The computation can be performed by summing throughout the enclosure.⁷³

VCP values are frequently associated with luminaires under standardized conditions of use. In this method, the luminaires are fractionally apportioned over the ceiling according to a standard scheme. VCP values are determined for^{70,75,76}:

- An initial average horizontal illuminance of 1,000 lx (approx. 90 fc)
- Room reflectances of $\rho_{cc} = 0.80$, $\rho_w = 0.50$, and $\rho_{fc} = 0.20$
- Luminaire mounting heights above the floor of 2.6, 3, 4, and 4.9 m (8.5, 10, 13, and 16 ft)
- A (given) range of room dimensions, including square, long narrow, and short wide rooms
- An observation point 1.2 m (4 ft) in front of the center of the rear wall and 1.2 m (4 ft) above the floor
- A horizontal line of sight, directly forward
- An upward limit to the field of view corresponding to an angle of 53° above and directly forward from the observer

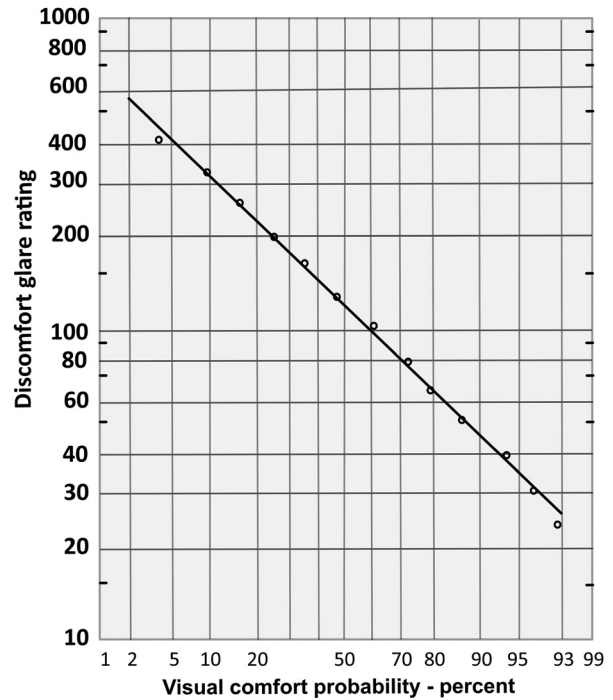


Figure A-13. A chart for converting discomfort glare ratings (DGR) to VCP (the percentage of observers expected to judge a given lighting condition to be either comfortable or at the borderline between comfort and discomfort).

There are two objectives in applying the standardized procedure. First, it simplifies calculations by permitting organization of various procedural steps. Second, it allows comparisons between those luminaires for which the standardized values have been tabulated even before a specific lighting layout has been made. **Figure A-14** illustrates an example of a tabulation of visual comfort probabilities as developed by the standardized procedure.

Example of a Tabulation of VCP Values									
Room		Luminaires Lengthwise				Luminaires Crosswise			
W	L	8.5	10.0	13.0	16.0	8.5	10.0	13.0	16.0
20	20	78	82	90	94	77	81	89	93
20	30	73	76	82	88	72	75	81	86
20	40	71	73	78	82	70	72	76	80
20	60	69	71	74	78	68	70	73	76
30	20	78	82	88	92	77	81	87	92
30	30	73	75	80	85	72	74	79	84
30	40	70	72	75	78	69	71	74	77
30	60	68	69	71	74	67	69	70	73
30	80	67	69	69	72	67	68	68	71
40	20	79	82	87	92	79	82	87	91
40	30	74	76	79	84	73	75	78	83
40	40	71	72	74	77	70	71	73	76
40	60	68	69	70	72	68	69	69	71
40	80	67	68	68	70	67	68	67	69
40	100	67	68	67	69	67	67	66	68
60	30	75	76	79	83	74	76	78	82
60	40	71	72	74	76	71	72	73	76
60	60	69	69	69	71	68	69	68	70
60	80	68	68	67	69	67	68	66	68
60	100	67	67	66	67	67	67	65	66
100	40	74	75	75	78	74	74	75	77
100	60	71	71	71	72	71	71	70	72
100	80	70	70	68	69	70	69	67	69
100	100	69	68	66	67	69	68	66	67

Figure A-14. Example of a tabulation of VCP values.

Annex B – Examples of Application of Radiative Transfer

B.1 Implementation of the Flux Balance Model

The simplest example of a radiative transfer problem solved by this method is a system with two surfaces. In this case, two equations in two unknowns result. The solution is:

$$M_1 = \frac{M_{01} + M_{02}\rho_1 f_{1 \rightarrow 2}}{1 - \rho_1 \rho_2 f_{1 \rightarrow 2} f_{2 \rightarrow 1}}, \quad (\text{B-1})$$

$$M_2 = \frac{M_{02} + M_{01}\rho_2 f_{2 \rightarrow 1}}{1 - \rho_1 \rho_2 f_{1 \rightarrow 2} f_{2 \rightarrow 1}}, \quad (\text{B-2})$$

where:

M_1 = final exitance of surface 1, taking account of interreflection

M_2 = final exitance of surface 2, taking account of interreflection

M_{01} = initial exitance of surface 1

M_{02} = initial exitance of surface 2

ρ_1 = diffuse reflectance of surface 1

ρ_2 = diffuse reflectance of surface 2

$f_{2 \rightarrow 1}$ = form factor from surface 2 to surface 1

$f_{1 \rightarrow 2}$ = form factor from surface 1 to surface 2

These equations provide a way to estimate the effects of interreflection in situations involving only two surfaces, or where only two elements of a large system participate in interreflection.

Another example of how a simple flux balance model can be applied to the analysis of a lighting system is the three-surface model used in the lumen method (see **Figure B-1**).⁷⁷⁻⁷⁹ The geometric arrangement is an empty rectangular room with recessed lighting equipment. The lighting system provides a general uniform illuminance on the work plane. Perfectly diffuse reflectances on all surfaces are assumed, and the same reflectance is assumed for the four wall surfaces. This allows the four walls to be treated as one surface. The average illuminance on the floor is to be calculated.

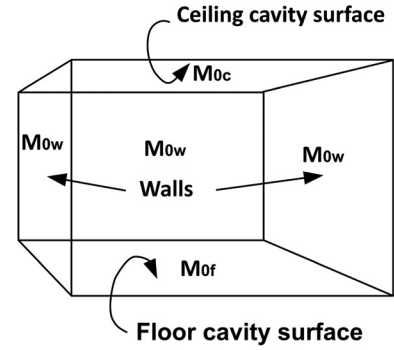


Figure B-1. Radiative transfer theory diagram. (© Illuminating Engineering Society)

For the purposes of the finite-element procedure, there are three elements or zones in this room: ceiling, walls, and floor. The surfaces have initial exitances M_{0c} , M_{0w} , and M_{0f} , respectively. These initial exitances are due to the flux falling onto the surfaces from the luminaires within the room. Since there are interreflections, the final exitances, M_c , M_w , and M_f , can be expressed as:

$$M_c = M_{0c} + \rho_c (M_w f_{c \rightarrow w} + M_f f_{c \rightarrow f}), \quad (\text{B-3})$$

$$M_w = M_{0w} + \rho_w (M_c f_{w \rightarrow c} + M_w f_{w \rightarrow w} + M_f f_{w \rightarrow f}), \quad (\text{B-4})$$

$$M_f = M_{0f} + \rho_f (M_c f_{f \rightarrow c} + M_w f_{f \rightarrow w}), \quad (\text{B-5})$$

where:

M_c = final ceiling exitance

M_w = final wall exitance

M_f = final floor exitance

M_{0c} = initial ceiling exitance

M_{0w} = initial wall exitance

M_{0f} = initial floor exitance

ρ_f = floor reflectance

ρ_w = wall reflectance

ρ_c = ceiling reflectance

$f_{f \rightarrow c}$ = form factor from floor to ceiling

$f_{f \rightarrow w}$ = form factor from floor to walls

$f_{w \rightarrow c}$ = form factor from walls to ceiling

$f_{w \rightarrow w}$ = form factor from walls to walls

$f_{w \rightarrow f}$ = form factor from walls to floor

$f_{c \rightarrow w}$ = form factor from ceiling to walls

$f_{c \rightarrow f}$ = form factor from ceiling to floor

With these three equations and the form factors, any interior illuminance or exitance can be determined. The above set of simultaneous equations is the basis for generating the tables of the coefficients of utilization, the wall exitance coefficients, and the ceiling cavity exitance coefficients.

This simple geometry permits the remaining six form factors to be determined from the form factor $f_{c \rightarrow f}$, between floor and ceiling, using reciprocity and the flux balance requirement (the sum of all form factors leaving a single surface sum to 1.0 for a closed system).

$$f_{f \rightarrow c} = f_{c \rightarrow f} \quad (\text{B-6})$$

$$f_{c \rightarrow w} = f_{f \rightarrow w} = 1 - f_{c \rightarrow f} \quad (\text{B-7})$$

$$f_{w \rightarrow c} = f_{w \rightarrow f} = \frac{A_c}{A_w} (1 - f_{c \rightarrow f}) \quad (\text{B-8})$$

$$f_{w \rightarrow w} = 1 - 2f_{w \rightarrow c} \quad (\text{B-9})$$

In these equations, A_c and A_w are the surface areas of the ceiling and walls, respectively. This three-surface model is the simplest of all flux transfer models for interior lighting calculations. It assumes that flux strikes each surface in a uniform manner and does not include variations in illuminance across any surface. In accordance with the International Commission on Illumination (CIE) practice and for the lumen method described below, the standard lighting coefficient tables are based upon a length-to-width ratio of 1.6 and given as a function of the room cavity ratio (RCR) only. **Table B-1** gives numerical values for the various form factors.

Table B-1. Form Factors for a Zonal Cavity System

RCR	$f_{w \rightarrow w}$	$f_{w \rightarrow c}$ $f_{w \rightarrow f}$	$f_{c \rightarrow w}$ $f_{f \rightarrow w}$	$f_{c \rightarrow f}$ $f_{f \rightarrow c}$
0	0.000	0.500	0.000	1.000
1	0.133	0.434	0.173	0.827
2	0.224	0.388	0.311	0.689
3	0.298	0.351	0.421	0.579
4	0.361	0.320	0.511	0.489
5	0.415	0.292	0.585	0.415
6	0.463	0.269	0.645	0.355
7	0.504	0.248	0.694	0.306
8	0.540	0.230	0.735	0.265
9	0.573	0.214	0.769	0.231
10	0.601	0.199	0.798	0.202

The lumen method extends the utility of this three-surface model by letting the ceiling and floor surfaces be virtual surfaces that are actually openings into rectangular cavities. These virtual surfaces are assigned effective reflectances. The top virtual surface is a plane containing the luminaires, and the cavity that this virtual surface represents extends from the luminaires up to the ceiling. The bottom virtual surface is the work plane, and the cavity that this virtual surface represents extends from the work plane down to the floor. It should be noted that the average illuminance onto the work plane is its final exitance divided by its effective reflectance.

Although more complicated, exactly the same procedure can be used for models of radiative transfer in a room that uses more than three surfaces (the finite-element method). Complex models can have thousands of discrete elements, as each surface in the room is divided into a collection of smaller zones.⁸⁰⁻⁸² The result is a more detailed knowledge of the room surface exitance distributions.

B.2 Monte Carlo Ray Tracing

One disadvantage of the simple flux balance models is that they can model reflections only from perfectly diffuse surfaces. Although this is adequate for most lighting calculations and architectural renderings, it does not provide sufficient information for estimating veiling reflections from specular surfaces and glare calculations.

Ray tracing techniques provide a more realistic model for the behavior of light, including reflection from specular surfaces and refraction by transparent objects. Reflections from diffuse surfaces can be approximated by reflecting hundreds of rays in random directions from each incident ray.⁸³ By providing appropriate weights to different directions of ray incidence and reflection, arbitrary bidirectional reflectance distribution functions can also be modeled.

Ray tracing is mathematically equivalent to radiative flux transfer between point emitters and receivers. Whereas evaluation of **Equation 5-3** (see **Section 5.2.1 Radiative Transfer**) provides estimates for the illuminance of each surface element, Monte Carlo ray tracing⁸⁴ provides similar estimates for random points on each surface. Illuminances for surface elements can then be obtained through Monte Carlo integration*

Rays can be traced either from the light sources into the environment (photosimulation)^{85,87} or from the observer into the environment (path tracing).^{86,87,89} Photosimulation methods are useful in that they provide accurate exitances for all surface elements in the environment. Path tracing methods are useful in that they provide accurate luminances for all surfaces within the observer's field of view, including veiling and specular reflections.

The disadvantage of ray tracing techniques is that millions to hundreds of millions of rays have to be evaluated in order to reduce the variance of the solutions to acceptable levels. While various statistical techniques can be used to ensure that randomly chosen ray directions transport on average the maximum amount of flux between surfaces,⁸⁷ convergence of the radiative transfer equation using ray tracing techniques requires modern hardware with multiple CPUs and/or threading techniques to achieve the speed needed in most use cases.

Annex C – Comparison of Calculated and Measured Quantities

Although it seems reasonable to expect calculated values of quantities to be reproduced in the field, in practice it is very difficult to reconcile measured quantities with those provided by calculation. Assumptions inherent in any calculation model often represent conditions very different from those in an actual lighting installation.⁸⁸ Some of these conditions are:

- Lambertian surfaces are assumed in most computer programs (see **Section 5.2.1 Radiative Transfer**, **Section 6.2.3 Radiative Transfer Calculations**, and **Section 10.1.3 Limitations**). Real-world surfaces may contain some degree of specularity.
- The room surface reflectance input to analysis software may not accurately represent what is present in the field.
- Reduced electrical voltage in the power system may produce reduced light output.
- The assumed ballast factor may be much different from that present in the field.
- Thermal effects in an installed luminaire may alter light output.
- Minor differences incurred in the manufacturing process or in the positioning of the lamp within a luminaire may alter the luminaire's photometric distribution.

* "Monte Carlo methods (or Monte Carlo experiments) are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. Their essential idea is using randomness to solve problems that might be deterministic in principle. They are often used in physical and mathematical problems and are most useful when it is difficult or impossible to use other approaches. Monte Carlo methods are mainly used in three problem classes: optimization, numerical integration, and generating draws from a probability distribution." Source: Wikipedia – Monte Carlo method: https://en.wikipedia.org/wiki/Monte_Carlo_method (Accessed 2018 Nov20).

Furniture and other absorbing and reflecting surfaces may not have been considered in the computer model. No analysis model is an exact representation of any real room. Simplifying assumptions in the calculation method may limit the accuracy of the results. Far-field photometric methods applied in a near-field situation may not accurately model the luminaire performance.

Another reason for disagreement can be errors in the measurement process. It is important to follow strict guidelines when measuring the photometric performance of lighting systems. For example, it is important that the lighting system be measured at a temperature that is representative of its thermal equilibrium condition. New lamps should operate for at least 100 hours before measurements are taken. It is critical that the operator of a photometer ensure that his or her own presence does not influence the reading. Orientation and positioning of the photometer are also critical; a tripod with a leveling device is particularly useful when conducting horizontal illuminance measurements. Finally, all daylight should be eliminated from measurements of interior electric lighting systems, perhaps by conducting measurements at night. Any attempt to subtract out daylight levels is difficult and is likely to introduce errors because of the temporal variations in daylight illuminance.

The magnitude of the differences between detailed analysis methods and field measurements varies. In general, differences of less than 20% can be expected, but in extreme cases, where a calculation method simply cannot handle the complexity of the lighting system, they may be greater. A more complete discussion of the uncertainties is available.⁸⁹

REFERENCES

- 1 DiLaura DL. A History of Light and Lighting. New York. Illuminating Engineering Society; 2006.
- 2 Illuminating Engineering Society. Software survey. *Light Des Appl.* 2005;35(9):69-77.
- 3 Ward GJ. The RADIANCE lighting simulation and rendering system. In: Proceedings of the 21st annual conference on computer graphics and interactive techniques. New York: ACM SIGGRAPH; 1994.
- 4 Sillion FX, Puech C. Radiosity and global illumination. San Francisco: Morgan Kaufmann; 1994.
- 5 Ashdown I. Radiosity: A programmer's perspective. New York: Wiley; 1994.
- 6 Lighting Technologies. Lumen Micro – User's Guide. Boulder, Colo.: Lighting Technologies; 1987.
- 7 Illuminating Engineering Society. ANSI/IES LS-2-20, Lighting Science: Concepts and Language of Lighting. New York: IES; 2020.
- 8 DiLaura DL, Quinlan J. Non-diffuse radiative transfer 1: Area sources and point receivers. *J Illum Eng Soc.* 1995;24(2):102-113.
- 9 Murdoch JB. Illuminating Engineering, 2nd ed. New York: Visual Communications; 2003.
- 10 Hamilton DC, Morgan WR. Radiant-interchange configuration factors. Washington, DC: National Advisory Committee for Aeronautics; 1952. (Technical Note 2836).
- 11 Siegel R, Howel JR. Thermal radiation heat transfer, 4th ed. New York: Taylor & Francis; 2002.
- 12 Gershun A. The light field, trans. *J Math Phys.* 1939;18(2):51-151.
- 13 Illuminating Engineering Society. ANSI/IES LM-72-20, Approved Method: Directional Positioning of Photometric Data. New York: IES; 2020.
- 14 Illuminating Engineering Society. ANSI/IES LS-4-20, Lighting Science: Measurement of Light – The Science of Photometry. New York: IES; 2020.
- 15 Yamauti Z. Geometrical calculation of illumination due to light from luminous sources of simple forms. Tokyo: Electrotechnical Laboratory; 1924. (Researches of the Electrotechnical Laboratory, 148).
- 16 Fock V. Zur Berechnung der Beleuchtungsstärke. *Z Phys.* 1924;28:102-113.
- 17 Illuminating Engineering Society. ANSI/IES LM-46-20, Approved Method: Photometric Testing of Indoor Luminaires Using High Intensity Discharge or Incandescent Filament Lamps. New York: IES; 2020.
- 18 Mistrick RG, English CR. A study of near-field indirect lighting calculations. *J Illum Eng Soc.* 1990;19(2):103-112.
- 19 Levin RE. Photometric characteristics of light controlling apparatus. *Illum. Eng.* 1971;66(4):205-215.
- 20 Lautzenheiser T, Weller G, Stannard S. Photometry for near field applications. *J Illum Eng Soc.* 1984;13(2):262-269.
- 21 Stannard S, Brass J. Application distance photometry. *J Illum Eng Soc.* 1990;18(1):39-46.
- 22 Ngai PY, Zhang JX, Zhang FG. Near-field photometry: Measurement and application for fluorescent luminaires. *J Illum Eng Soc.* 1992;21(2):68-83.
- 23 Ashdown I. Near-Field Photometry: A new approach. *J Illum Eng Soc.* 1993;22(1):163-180.
- 24 Ashdown I. Making near-field photometry practical. *J Illum Eng Soc.* 1998;27(1):67-79.

- 25 Illuminating Engineering Society. ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 Jul 19).
- 26 Murray-Coleman JF, Smith AM. The automated measurement of BRDFs and their application to luminaire modeling. *J Illum Eng Soc*. 1990;19(1):87-99.
- 27 Ward GJ. Measuring and modeling anisotropic reflection. *Comp Graphics*. 1992;26(2):265-272.
- 28 Illuminating Engineering Society. ANSI/IES RP-8-18, Recommended Practice for Design and Maintenance of Roadway and Parking Facility Lighting. New York: IES; 2018.
- 29 Ashdown I. Color and Reflectance. Published online: <https://lightinganalysts.freshdesk.com/support/solutions/articles/22000209131-info-color-and-reflectance>. (Accessed 2018 Nov 5).
- 30 Illuminating Engineering Society. ANSI/IES LS-3-20, Lighting Science: Physics and Optics of Radiant Power. New York: IES; 2020.
- 31 DeBoer J. Modeling indoor illumination by complex fenestration systems based on bidirectional photometric data. *Energy Build*. 2006;38(7):849-868.
- 32 Maamari F, Andersen M, de Boer J, Carroll W, Dumortier D, Greenup P. Experimental validation of simulation methods for bi-directional transmission properties at the daylighting performance level. *Energy and Buildings*. 2006;38(7):218.
- 33 Illuminating Engineering Society. ANSI/IES LM-63-19, IES Standard File Format for the Electronic Transfer of Photometric Data and Related Information. New York: IES; 2019.
- 34 Illuminating Engineering Society. ANSI/IES LP-3-20, Lighting Practice: Designing and Specifying Daylighting for Buildings. New York: IES; 2020.
- 35 Cohen MF, Wallace JR. Radiosity and realistic image synthesis. Boston: Academic Press Professional; 1993.
- 36 Dutré P. Advanced Global Illumination. Natick, Mass.: A K Peters; 2006.
- 37 Freniere ER, Gregory GG, Hassler RA. Polarization models for Monte Carlo ray tracing. In: *Proc SPIE*. 1999;3780(22):148150.
- 38 Jensen HW. Realistic Image Synthesis Using Photon Mapping. Natick, Mass.: A K Peters; 2001.
- 39 Ashdown I. Radiosity: A Programmer's Perspective. New York: Wiley; 1994.
- 40 Pharr M, Humphreys G. Physically Based Rendering. Morgan Kaufmann; 2004.
- 41 Illuminating Engineering Society. ANSI/IES LS-5-20, Lighting Science: Color. New York: IES; 2020.
- 42 Birn J. Digital Lighting & Rendering. San Francisco: New Riders; 2000.
- 43 Kaneko T. Detailed shape representation with parallax mapping. In: *Proc ICAT*; 2001:205-8.
- 44 Ruppertsberg AI, Bloj M. Rendering complete scenes for psychophysics using RADIANCE: How accurate can you get? *J Opt Soc Am A*. 2006;23(4):759-68.
- 45 Seetzen H, Whitehead L, Ward G. A high dynamic range display system using low and high resolution modulators. In: *Proceedings of the Society for Information Display Symposium*; 2003.
- 46 Ledda P, Chalmers A, Troschianko T, Seetzen H. Evaluation of tone mapping operators using a high dynamic range display. *ACM Trans Graphics*. 2005;24(3):640-8.

- 47 Johnson G, Fairchild M. 2003. Rendering HDR images. In: Proceedings of Imaging Science and Technology & Society for Information Display; 2003:36-41.
- 48 Reinhard E, Ward G, Pattanaik S, Debevec P. High Dynamic Range Imaging. San Francisco: Morgan Kaufman; 2006.
- 49 International Commission on Illumination (CIE). CIE 171:2006, Test cases to assess the accuracy of lighting computer programs. Vienna CIE; 2006.
- 50 Maamari F, Fontoynt, Adra N. 2006. Application of the CIE test cases to assess the accuracy of lighting computer programs. Energy and Build. 2006;38:869-77.
- 51 Slater A, Graves H. Benchmarking lighting design software. London: CIBSE; 2002. (CIBSE TM 28).
- 52 Geisler-Moroder D, Dür A. Validation of Radiance against CIE 171:2006. In: 7th International Radiance Workshop. Switzerland; 2008.
- 53 Dau Design Consulting. Validation of AGI32 against CIE 171:2006. Calgary: Dau Design Consulting; 2007.
- 54 Laouadi A, Arenault C. Validation of skylight performance assessment software. ASHRAE Trans; 2006 Jul.
- 55 Illuminating Engineering Society. ANSI/IES RP-36-20, Recommended Practice: Lighting Maintenance. New York: IES; 2020.
- 56 Eble-Hankins M, Waters CE. Subjective impression of discomfort glare from sources of non-uniform luminance. Leukos. 2009;6(1):51-77.
- 57 Siegel R. Thermal Radiation Heat Transfer. 5th ed. Boca Raton, FL: CRC Press; 2010.
- 58 Illuminating Engineering Society. Recommended practice for classification of interior luminaires by distribution: Luminaire spacing criterion. Light Des Appl. 1977;7(8):20-21.
- 59 Rea MS and Ouellette MJ. Relative visual performance: A basis for application. Lighting Res Techn. 1991;23(3):135-44.
- 60 Rea MS and Ouellette MJ. Visual performance using reaction times. Lighting Res Techn. 1988;20(4):139-53.
- 61 Luckiesh M, Guth SK. Brightness in visual field at borderline between comfort and discomfort (BCD). Illum Eng. 1949;44(11):650-70.
- 62 Hopkinson RG. Evaluation of glare. Illum Eng. 1957;52(6): 305-16
- 63 Guth SK, McNelis JF. Further data on discomfort glare from multiple sources. Illum Eng. 1961;56(1):46-57.
- 64 Guth SK. A method for the evaluation of discomfort glare. Illum Eng. 1963;57(5):351-64.
- 65 Bradley RD, Logan HL. A uniform method for computing the probability of comfort response in a visual field. Illum Eng. 1964;59(3):189-206.
- 66 Allphin W. Further studies of sight line and direct discomfort glare. Illum Eng. 1968;63(1):26-31.
- 67 Allphin W. BCD appraisals of luminaire brightness in a simulated office. Illum Eng. 1961;56(1):31-44.
- 68 Allphin W. Further appraisals of luminaire brightness. Illum Eng. 1961;56(12):701-7.
- 69 McGowan TK, Guth SK. Extending and applying the IES visual comfort rating procedure. Illum Eng. 1969;64(4):253-70.
- 70 Illuminating Engineering Society. Outline of a standard procedure for computing visual comfort ratings for interior lighting. J Illum Eng Soc. 1973;2(3):328-44. (IES RQQ Report No. 2, 1972).

- 71 Fry GA. A simplified formula for discomfort glare. *J Illum Eng Soc.* 1976;8(1):10-20.
- 72 Levin RE. An evaluation of VCP calculations. *J Illum Eng Soc.* 1973;2(4):355-61.
- 73 DiLaura DL. On the computation of visual comfort probability. *J Illum Eng Soc.* 1976;5(4):207-17.
- 74 Levin RE. Position index in VCP calculations. *J Illum Eng. Soc.* 1975;4(2):99-105.
- 75 Illuminating Engineering Society. Outline of a standard procedure for computing visual comfort ratios for interior lighting. *Illum Eng.* 1966;61(10):643-66. (IES RQQ Report No. 2.)
- 76 Illuminating Engineering Society. Committee on Recommendations of Quality and Quantity of Illumination. A statement concerning visual comfort probability (VCP): Naive vs experienced observers. *Illum Eng.* 1969;64(9):604.
- 77 O'Brien PF, Balogh E. Configuration factors for computing illumination within interiors. *Illum Eng.* 1867;62(4): 169-79.
- 78 Illuminating Engineering Society. Zonal-cavity method of calculating and using coefficients of utilization. *Illum Eng.* 1964;59(5):309-28.
- 79 Jones JR, Jones BF. Using the zonal-cavity system in lighting calculations: Part I: *Illum. Eng.* 1964;59(5):413–415; Part II: 1964;59(6):448-50; Part III: 1964;59(7):501-3; Part IV: 1964;59(8):556-61.
- 80 Mistrick RG, DiLaura DL. A new finite orthogonal transform applied to radiative transfer calculations. *J Illum Eng Soc.* 1987;16(2):115-28.
- 81 Mistrick RG. A priority based dual density finite element interreflected component calculation. *J Illum Eng Soc.* 1989;18(2):16-22.
- 82 Zhang JX, Ngai PY. Lighting calculations in a multi-partitioned space. *J Illum Eng Soc.* 1991;20(1):32-43.
- 83 Sillion F, Puech C. Radiosity and global illumination. San Francisco: Morgan Kaufmann; 1994.
- 84 Shirley P, Wang C, Zimmerman K. Monte Carlo techniques for direct lighting calculations. *ACM Trans Computer Graphics.* 1996;15(1):1-36.
- 85 Ward Larson G, Shakespeare RA. Rendering with radiance: The art and science of lighting visualization. San Francisco: Morgan Kaufmann; 1988.
- 86 Urena C., Pueyo X, Torres JC. A formalization and classification of global illumination methods. *Computer Graphics.* 1997;21(2):225–236.
- 87 Ward GJ, Rubinstein FM. A ray tracing solution for diffuse interreflection. *Computer Graphics.* 1988;22(4):85-92.
- 88 Levin RE. The photometric connection. Part 1: *Light Des Appl.* 1982;12(9):28–35; Part 2: 1982;12(10): 60-3; Part 3: 1982;12(11):42–47; Part 4: 1982;12(12):16-18.
- 89 Illuminating Engineering Society. Calculation of luminance coefficients based upon the zonal-cavity method. *Illum. Eng.* 1968;63(8):423-32.



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